Chapter 1
Introduction



Abstract In this introductory chapter, we sketch the existing work in machine learning on which we build and then outline the contents of the book.

Keywords Machine learning · Conformal prediction · Venn prediction · Assumption of randomness

1.1 Machine Learning

The rapid development of computer technology during the last several decades has made it possible to solve ever more difficult problems in a wide variety of fields. The development of software has been essential to this progress. The painstaking programming in machine code or assembly language that was once required to solve even simple problems has been replaced by programming in high-level object-oriented languages. We are concerned with the next natural step is this progression—the development of programs that can *learn*, i.e., automatically improve their performance with experience.

The need for programs that can learn was already recognized by Alan Turing [358] in 1950, who argued that it may be too ambitious to write from scratch programs for tasks that even humans must learn to perform. Consider, for example, the problem of recognizing hand-written digits. We are not born able to perform this task, but we learn to do it quite robustly. Even when the hand-written digit is represented as a grey-scale matrix, as in Fig. 1.1, we can recognize it easily, and our ability to do so scarcely diminishes when it is slightly rotated or otherwise perturbed. We do not know how to write instructions for a computer that will produce equally robust performance.

The essential difference between a program that implements instructions for a particular task and a program that learns is adaptability. A single learning program may be able to learn a wide variety of tasks: recognizing hand-written digits and faces, diagnosing patients in a hospital, estimating house prices, etc.

Recognition, diagnosis, and estimation can all be thought of as special cases of prediction. A person or a computer is given certain information and asked to

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Fig. 1.1 A hand-written digit



predict the answer to a question. A broad discussion of learning would go beyond prediction to consider the problems faced by a robot, who needs to act as well as predict. The literature on machine learning, has emphasized prediction, however, and the research reported in this book is in that tradition. We are mostly interested in algorithms that learn to predict well.

1.1.1 Learning Under Randomness

One learns from experience. This is as true for a computer as it is for a human being. In order for there to be something to learn there must be some stability in the environment; it must be governed by constant, or evolving only slowly, laws. And when we learn to predict well, we may claim to have learned something about that environment.

The traditional way of making the idea of a stable environment precise is to assume that it generates a sequence of examples randomly from some fixed probability distribution, say Q, on a fixed space of possible examples, say $\mathbf{Z}$. These mathematical objects, $\mathbf{Z}$ and Q, describe the environment.

The environment can be very complex; $\mathbf{Z}$ can be large and structured in a complex way. This is illustrated by the USPS dataset from which Fig. 1.1 is drawn (see Sect. B.1 in Appendix B). Here an example is any 16×16 image with 31 shades of grey, together with the digit the image represents (an integer between 0 to 9). So there are $31^{16\times16} \times 10$ (this is approximately 10^{382}) possible examples in the space $\mathbf{Z}$.

In most of this book, we assume that each example consists of an *object* and its *label*. In the USPS dataset, for example, an object is a grey-scale matrix like the one in Fig. 1.1, and its label is the integer between 0 and 9 represented by the grey-scale matrix.

In the problem of recognizing hand-written digits and other typical machine-learning problems, it is the space of objects, the space of possible grey-scale images, that is large. The space of labels is either a small finite set (in what is called *classification problems*) or the set of real numbers (*regression problems*).

When we say that the examples are chosen randomly from Q, we mean that they are independent in the sense of probability theory and all have the distribution Q. They are independent and identically distributed (IID). We call this the *randomness assumption* (or, sometimes, statistical randomness assumption, when we want to emphasize that it is not related to Kolmogorov's algorithmic randomness).

Of course, not all work in machine learning is concerned with learning under randomness. In learning with expert advice, for example, randomness is replaced

by a game-theoretic set-up [45, 381]; here a typical result is that the learner can predict almost as well as the best strategy in a pool of possible strategies. In reinforcement learning, which is concerned with rational decision-making in a dynamic environment [351], the standard assumption is Markovian. In this book, we will consider extensions of learning under randomness in Parts III and IV.

1.1.2 Learning Under Unconstrained Randomness

Sometimes we make the randomness assumption without assuming anything more about the environment: we know the space of examples $\mathbf{Z}$, we know that examples are drawn independently from the same distribution, and this is all we know. We know nothing at the outset about the probability distribution Q from which each example is drawn. In this case, we say we are *learning under unconstrained randomness*. Most of the work in this book, like much other work in machine learning, is concerned with learning under unconstrained randomness.

The strength of modern machine-learning methods often lies in their ability to make hedged predictions under unconstrained randomness in a *high-dimensional* environment, where examples have a very large (or infinite) number of components. We already mentioned the USPS dataset, where each example consists of 257 components (16×16 pixels and the label). In machine learning, this number is now considered small, and the problem of learning from the USPS dataset is sometimes regarded as a toy problem.

1.2 A Shortcoming of Statistical Learning Theory

Machine learning has made significant strides in its study of learning under unconstrained randomness. We now have a wide range of algorithms that often work very well in practice: neural networks, decision trees, nearest neighbours algorithms, and naive Bayes methods have been used for decades; newer algorithms include support vector machines, random forests, and boosting, and in recent years neural networks have again become extremely popular in the form of deep learning.

From a theoretical point of view, traditional machine learning's most significant contributions to learning under unconstrained randomness fall under the umbrella of *statistical learning theory*. This theory, which began with the discovery of VC dimension by Vapnik and Chervonenkis in the late 1960s and was partially rediscovered independently by Valiant [359] in 1984, has produced both deep mathematical results and learning algorithms that work well in practice (see Vapnik [366] for a review).

Given a "training" set of examples, a prediction algorithm of machine learning produces what we call a *prediction rule*—a function mapping the objects into the labels. This is illustrated in Fig. 1.4 (where "prediction rule" is referred to as

"general rule"). Usually, the value taken by a prediction rule on a new object is a *simple prediction*—a guess that is not accompanied by any statement concerning how accurate it is likely to be. Statistical learning theory does guarantee, however, that, under some conditions on the prediction algorithm, as the training set becomes bigger and bigger these predictions will become more and more accurate with greater and greater probability: they are *probably approximately correct*.

How probably and how approximately? This question has not been answered as well as we might like by statistical learning theory. This is because the theoretical results that might be thought to answer it, the bounds that demonstrate arbitrarily good accuracy with sufficiently large sizes of the training set, are usually too loose to tell us anything interesting for training sets that we actually have. This happens in spite of the empirical fact that the predictions often perform very well in practice. Consider, for example, the problem of recognizing hand-written digits, which we have already discussed. Here we are interested in giving an upper bound on the probability that our learning algorithm fails to choose the right digit; we might like this probability to be less than 0.05, for example, so that we can be 95% confident that the prediction is correct. Unfortunately, typical upper bounds on the probability of error provided by the theory, even for relatively clean datasets such as the USPS dataset we have discussed, are greater than 1; bounds less than 1 can usually be achieved only for very straightforward problems or with very large datasets. This is true even for newer results in which the bound on the accuracy depends on the training set (cf. Sect. 13.2.3).

1.2.1 The Hold-Out Estimate of Confidence

Fortunately, there are less theoretical and more effective ways of estimating the confidence we should have in predictions output by machine-learning algorithms. One of the most effective is the oldest and most naive: the "hold-out" estimate. In order to compute this estimate, we split the available examples into two parts, a training set and a hold-out sample, playing the role of a test set. We apply the algorithm to the training set in order to find a prediction rule, and then we apply this prediction rule to the hold-out sample. The observed rate of errors on the hold-out sample tells us how confident we should be in the prediction rule when we apply it to new examples (for details, see Sect. 1.2.1).

1.2.2 The Contribution of This Book

When we use a hold-out sample to obtain a bound on the probability of error, or when we use an error bound from statistical learning theory, we are *hedging* the prediction—we are adding to it a statement about how strongly we believe it. In this book, we develop a different way of producing hedged predictions. Aside from the



Fig. 1.2 In the problem of digit recognition, we would like to attach lower confidence to the prediction for the image in the middle than to the predictions for the images on the left and the right

elegance of our new methods, at least in comparison with the procedure that relies on a hold-out sample, the methods we develop have several important advantages.

As mentioned in the preface to the first edition, we do not always have the rigid separation between learning and prediction, which is the feature of the traditional approaches that makes hedged prediction feasible. In our basic learning protocol learning and prediction are blended, yet our predictions are hedged.

Second, the hedged predictions produced by our new algorithms are much more confident and accurate. We have, of course, a different notion of a hedged prediction, so the comparison can be only informal; but the difference is so big that there is little doubt that the improvement is real from the practical point of view.

A third advantage of our methods is that the confidence with which the label of a new object is predicted is always tailored not only to the previously seen examples but also to that object (see Fig. 1.2).

1.3 The Online Framework

The new methods presented in this book are quite general; they can be tried out, at least, in almost any problem of learning under randomness. The framework in which we introduce, and then study, these methods is somewhat unusual, however. Most previous theoretical work in machine learning has been in an *offline* framework: one uses a batch of old examples to produce a prediction rule, which is then applied to new examples. We begin instead with a framework that is *online*: one makes predictions sequentially, basing each new prediction on all the previous examples instead of repeatedly using a rule constructed from a fixed batch of examples.

1.3.1 Online Learning

Our basic framework is *online* because we assume that the examples are presented one by one. Each time, we observe the object and predict the label. Then we observe the label and go on to the next example. We start by observing the first object x_1 and predicting its label y_1. Then we observe y_1 and the second object x_2, and predict its label y_2. And so on. At the nth step, we have observed the previous examples

$$(x_1, y_1), \ldots, (x_{n-1}, y_{n-1}) \tag{1.1}$$

and the new object x_n, and our task is to predict y_n. The quality of our predictions should improve as we accumulate more and more old examples. This is the sense in which we are learning.

1.3.2 Online/Offline Compromises

The methods we present in this book are most naturally described and are most amenable to mathematical analysis in the online framework. However, they also extend to relaxations of the online protocol that make it close to the offline setting, and this is important, because most practical problems have at least some offline aspects. If we are concerned with recognizing hand-written zip codes, for example, we cannot always rely on a human teacher to tell us the correct interpretation of each hand-written zip code; why not use such an *ideal teacher* directly for prediction? The relaxation of the online protocol considered in Sect. 3.3 includes "slow teachers", who provide the feedback with a delay, and "lazy teachers", who provide feedback only occasionally. In the example of zip codes recognition, this relaxation allows us to replace constant supervision by using returned letters for teaching or by occasional lessons.

1.3.3 One-Off and Offline Learning

We will sometimes discuss prediction algorithms in the simplest possible setting, which we will refer to as *one-off learning*. A one-off predictor is given a training set (1.1) and a test object x_n; its task is to predict the label y_n of x_n. It is applied only once.

A one-off predictor can be embedded into both online and offline frameworks (see Fig. 1.3). In the online framework, new examples arrive sequentially, their labels are predicted given their objects, and they are added to the training set, as described earlier.

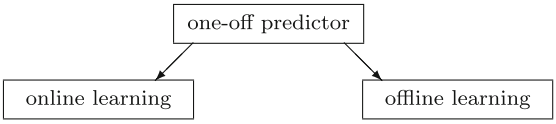


Fig. 1.3 One-off, online, and offline learning: two ways of embedding one-off predictors into a bigger learning protocol

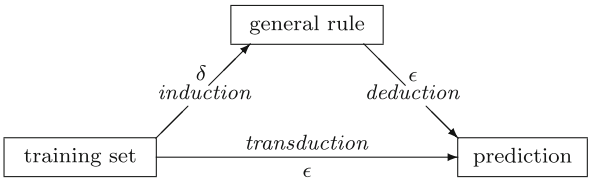


Fig. 1.4 Inductive and transductive (or eductive) prediction

In the offline framework (and sometimes in the one-off framework), there is a shift in our notation. While in the one-off and online settings we usually write $z_1, \dots, z_{n-1}$ for the old examples and z_n for the new example (especially in the online setting), now that we are also interested in the case where the same rule may be applied to many new examples, we usually write $z_1, \dots, z_l$ for the old examples and $z_{l+1}, \dots, z_{l+k}$ for the new examples. The one-off predictor is applied independently k times to the old examples $z_1, \dots, z_l$ and the new object x_i, $i = l+1, \dots, l+k$, to predict its label y_i.

1.3.4 Induction, Transduction, and the Online Framework

Vapnik's [366] distinction between induction and transduction, as applied to the problem of prediction, is depicted in Fig. 1.4. In *inductive prediction* we first move from examples in hand to some more or less general rule, which we might call a prediction or decision rule, a model, or a theory; this is the *inductive step*. When presented with a new object, or a test set of new objects, we derive a prediction from the general rule; this is the *deductive step*. In *transductive prediction*, we take a shortcut, moving from the old examples directly to the prediction(s) about the new object(s).

In philosophy transduction had been known as eduction since its introduction by W. E. Johnson in 1924 in Part III [182, Chap. IV] of his Magnum Opus *Logic* [180–182]. He points out that the notion (unlike the term) of eduction is well-known, being "the kind of inference which Mill speaks of as from particulars to particulars" (Johnson prefers "inference from instances to instances", which in our terminology becomes "inference from examples to examples"). In the same book Johnson introduced the notion of exchangeability and used it as an assumption for eduction. We will continue the discussion of exchangeability in Sect. 2.9.1.

Typical examples of the inductive step are estimating parameters in statistics and finding a "concept" (to use Valiant's [359] terminology) in statistical learning theory. Examples of transductive prediction are estimation of future observations in statistics (see, e.g., [64, Sect. 7.5]) and nearest neighbours algorithms in machine learning.

In machine learning, transduction is usually discussed in the case where the test set contains a significant number of new objects, whereas in this book we are often

William Ernest Johnson (1858–1931).

The photograph is in public domain



interested in the case of a single new object. When this is not clear from the context, we will refer to this case as *single-object transduction*.

In the case of simple predictions, the distinction between induction and single-object transduction is less than crisp. A method for doing single-object transduction is a method for predicting the label y of a new object x from a training set $x_1, y_1, \ldots, x_l, y_l$ and x itself. Such a method gives a prediction for any object that might be presented as x, and so it defines, at least implicitly, a rule, which might be extracted from $x_1, y_1, \ldots, x_l, y_l$ (induction), stored, and then subsequently applied to x to predict y (deduction). So any real distinction is really at a practical and computational level: do we extract and store the general rule or not?

For hedged predictions the difference between transduction and induction goes deeper. We will typically want different notions of hedged prediction in the two frameworks. Mathematical results about induction typically involve two parameters, often denoted ϵ (the desired accuracy of the prediction rule) and δ (the probability we are willing to tolerate of failing to achieve the accuracy of ϵ), whereas results about transduction involve only one parameter, which we will denote ϵ (in this book, the probability of error we are willing to tolerate); see Fig. 1.4. A detailed discussion can be found in Chap. 13, which also contains a historical perspective on the three main approaches to hedged prediction (inductive, transductive, and Bayesian).

When we work in the online protocol, we would want to use a general rule extracted from $x_1, y_1, \ldots, x_{n-1}, y_{n-1}$ only once, to predict y_n from x_n. After observing x_n and then y_n, we have a larger dataset, $x_1, y_1, \ldots, x_n, y_n$, and we can use it to extract a new, possibly improved, general rule before trying to predict y_{n+1} from x_{n+1}. So from a purely conceptual point of view, induction seems silly in the online framework; it is more natural to say that we are doing transduction, even in cases where the general rule is easy to extract. As a practical matter, however, the computational cost of a transductive method may be high, and in this case, it may be sensible to compromise with the inductive approach. After accumulating a certain number of examples, we might extract a general rule and use it for a while, only updating it as frequently as is practical.

1.4 Conformal Prediction

Most of this book is devoted to a particular method that we call "conformal prediction". When we use this method, we predict that a new object will have a label that makes it similar to the old examples in some specified way, and we use the degree to which the specified type of similarity holds within the old examples to estimate our confidence in the prediction. Our conformal predictors are, in other words, "confidence predictors".

We need not explain here exactly how conformal prediction works. This is the topic of the next chapter. But we will explain informally what a confidence predictor aims to do and what it means for it to be valid and efficient.

1.4.1 Nested Prediction Sets

Suppose we want to pinpoint a target that lies somewhere within a rectangular field. This could be an online prediction problem; for each example, we predict the coordinates $y_n \in [a_1, a_2] \times [b_1, b_2]$ of the target from a set of measurements x_n.

We can hardly hope to predict the coordinates y_n exactly. But we can hope to have a method that gives a subset Γ_n of $[a_1, a_2] \times [b_1, b_2]$ where we can be confident y_n lies. Intuitively, the size of the *prediction set* Γ_n should depend on how great a probability of error we want to allow, and in order to get a clear picture, we should specify several such probabilities. We might, for example, specify the probabilities 1%, 5%, and 20%, corresponding to *confidence levels* 99%, 95%, and 80%. When the probability of the prediction set failing to include y_n is only 1%, we declare 99% confidence in the set (highly confident prediction). When it is 5%, we declare 95% confidence (confident prediction). When it is 20%, we declare 80% confidence (casual prediction). We might also want a 100% confidence set, but in practice this might be the whole field assumed at the outset to contain the target.

Figure 1.5 shows how such a family of prediction sets might look. The casual prediction pinpoints the target quite well, but we know that this kind of prediction can be wrong 20% of the time. The confident prediction is much bigger. If we want

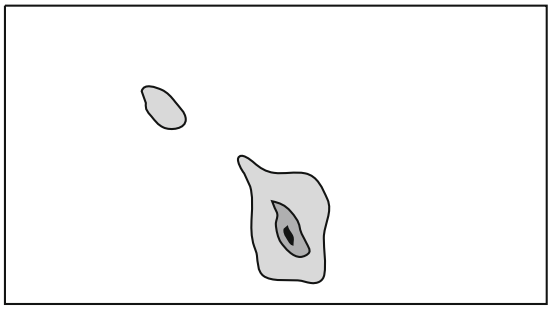


Fig. 1.5 An example of a nested family of prediction sets (casual prediction in black, confident prediction in dark grey, and highly confident prediction in light grey)

to be highly confident (make a mistake only for each 100th example, on average), we must accept an even lower accuracy; there is even a completely different location that we cannot rule out at this level of confidence. In principle, a confidence predictor outputs prediction sets for all confidence levels, and these sets are nested, as in Fig. 1.5.

There are two important desiderata for confidence predictors:

- They should be *valid*, in the sense that in the long run the frequency[1] of error does not exceed ϵ at each chosen confidence level $1 - \epsilon$.
- They should be *efficient*, in the sense that the prediction sets they output are as small as possible.

We would also like the predictor to be as conditional as possible—e.g., we want it to take full account of how difficult the particular example is.

1.4.2 Validity

Our conformal predictors are always valid (although we have different kinds of validity for different varieties of conformal predictors; here we concentrate on the basic variety, full conformal predictors). Figure 1.6 confirms empirically the validity for one particular conformal predictor that we study in Chap. 3. The solid, dash-dot, and dotted lines show the cumulative number of errors for the confidence levels 99%, 95%, and 80%, respectively; the value of each graph at n is the total number of errors made over the first n examples. As expected, the number of errors made grows linearly, and the slope is approximately 20% for the confidence level 80%, 5% for the confidence level 95%, and 1% for the confidence level 99%. Some details of our computational experiments will be described in Appendix B (Sect. B.5).

As we will see in Chap. 2, a precise discussion of the validity of conformal predictors actually requires that we distinguish two kinds of validity: conservative and exact. In general, a conformal predictor is conservatively valid: the probability it makes an error when it outputs a prediction set at a confidence level $1 - \epsilon$ is no greater than ϵ, and there is little dependence between errors it makes when predicting successive examples (at successive *trials*, or *steps*, as we will say). This implies, by the law of large numbers, that the long-run frequency of errors at confidence level $1 - \epsilon$ is about ϵ or less. In practice, the conservativeness is often not great, especially when n is large, and so we get empirical results like those in Fig. 1.6, where the long-run frequency of errors is close to ϵ. From a theoretical point of view, however, we must introduce a small element of deliberate randomization into the prediction process in order to get exact validity, where the probability of a prediction set at a confidence level $1 - \epsilon$ being in error is exactly ϵ, errors are made independently at different trials, and the long-run frequency of errors converges to ϵ.

[1] By "frequency" we usually mean "relative frequency".

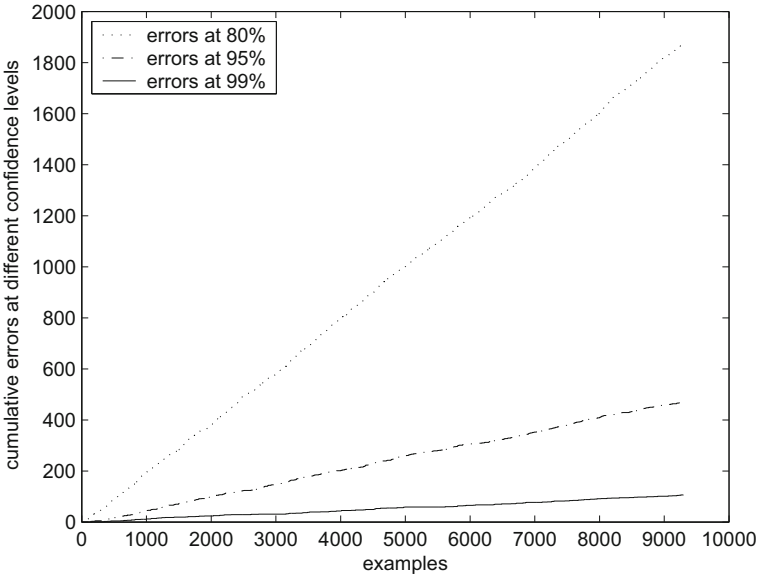


Fig. 1.6 Online performance of a conformal predictor ("the 1-nearest neighbour conformal predictor", described in Sect. 3.2.1) on the USPS dataset (9298 hand-written digits, randomly permuted) for the confidence levels 80, 95, and 99%. The figures in this book are not too much affected by statistical variation (due to the random choice of the permutation of the dataset)

1.4.3 Efficiency

Machine learning has been mainly concerned with two types of problems:

- Classification, where the label space $\mathbf{Y}$ is a small finite set (often binary).
- Regression, where the label space is the real line (or, sometimes, its simple subset, such as the positive real numbers).

In regression problems, the prediction set output by a conformal predictor is often an interval of values, and a natural measure of efficiency of such a prediction is the length of the interval, at different confidence levels. In classification problems, a natural measure of efficiency is the size of the prediction set—the number of labels in it, perhaps apart from the true label (which will be in the prediction set with high probability for a valid predictor). Measuring efficiency in the case of classification will be one of the major topics of Chap. 3.

As we will see in Chaps. 2 and 3, there are many conformal predictors for any particular online prediction problem, whether it is a classification problem or a regression problem. Indeed, we can construct a conformal predictor from any method for scoring the similarity (conformity, as we call it) of a new example to old ones. All these conformal predictors, it turns out, are valid. Which is most efficient—which produces the smallest prediction sets in practice—will depend on details of the environment that we may not know in advance.

1.4.4 Conditionality

The goal of conditionality can be explained with a simple example discussed by David Cox [62]. Suppose there are two categories of objects, "easy" (easy to predict) and "hard" (hard to predict). We can tell which objects belong to which category, and the two categories occur with equal probability; about 50% of the objects we encounter are easy, and 50% hard. We have a prediction method that applies to all objects, hard and easy, and has error rate 5%. We do not know what the error rate is for hard objects, but perhaps it is 8%, and we get an overall error rate of 5% only because the rate for easy objects is 2%. In this situation, we may feel uncomfortable, when we encounter a hard object, about appealing to the average error rate of 5% and saying that we are 95% confident of our prediction.

Whenever there are features of objects that we know make the prediction easier or harder, we would like to take these features into account—to condition on them. This is done by conformal predictors almost automatically: they are designed for specific applications so that their predictions take fullest possible account of the individual object to be predicted. What is not achieved automatically is the validity separately for hard and easy objects. It is possible, for example, that if a figure such as Fig. 1.6 were constructed for easy objects only, or for hard objects only, the slopes of the cumulative error lines would be different. We would get the correct slope if we average the slope for easy objects and the slope for hard objects, but we would ideally like to have the "conditional validity": validity for both categories of objects. As we show in Sect. 4.6, this can be achieved by modifying the definition of conformal predictors. In fact, the conditional validity is handled by a general theory that also applies when we segregate examples not by their difficulty but by their time of arrival, as when we are using an inductive rule that we update only at specified intervals.

1.4.5 Flexibility of Conformal Predictors

A useful feature of our method is that a conformal predictor can be built on top of almost any machine-learning algorithm. The latter, which we call the *underlying algorithm*, may produce hedged predictions, simple predictions, or simple predictions complemented by ad hoc measures of confidence; our experience is that it is always possible to transform it into a conformal predictor that inherits its predictive performance but is, of course, valid, just like any other conformal predictor. In this book we explain how to build conformal predictors using such methods as nearest neighbours, support vector machines, bootstrap, boosting, neural networks, decision trees, random forests, ridge regression, logistic regression, and any Bayesian algorithm (see Sects. 2.3, 2.4, 3.2, and 4.3).

1.5 Probabilistic Prediction Under Unconstrained Randomness

There are many ways to do classification and regression under unconstrained randomness and for high-dimensional examples. Conformal predictors, for example, combine good theoretical properties with high accuracy in practical problems. It is true that the environment has to be benign, in some sense, for any learning method to be successful, but there are no obvious insurmountable barriers for classification and regression. The situation changes if we move to the harder problem of *probabilistic prediction*: that of guessing the probability distribution for the new object's label. Features of data that can reasonably be expected in typical machine-learning applications become such barriers.

For simplicity, we will assume in this section, apart from the last short subsection, that the label is binary, 0 or 1. In this case the probabilistic prediction for the label of the new object boils down to one number, the predicted probability that the label is 1.

The problem of probabilistic prediction is discussed in Chaps. 5–7 and 12–13. Probabilistic prediction is impossible in an important sense, but there are also senses in which it is possible. So this book gives more than one answer to the question "Is probabilistic prediction possible?" We start with a "yes" answer.

1.5.1 Universally Consistent Probabilistic Predictor

Stone [347] showed that a nearest neighbours probabilistic predictor (whose probabilistic prediction is the fraction of objects classified as 1 among the k nearest neighbours of the new object, with a suitably chosen k) is *universally consistent*, in the sense that the difference between the probabilistic prediction and the true conditional probability given the object that the label is 1 converges to zero in probability. The only essential assumption is randomness;[2] there are no restrictive regularity conditions.

Stone's actual result was more general, and it has been further extended in different directions. We will state similar results in Chaps. 6 and 7 in the context of conformal prediction.

[2] The other assumption made by Stone was that the objects were coming from a Euclidean space; since "Euclidean" is equivalent to "standard Borel" in the context of existence of a universally consistent probabilistic predictor, this assumption is very weak. See Sect. A.1.1 for further information about standard Borel spaces.

1.5.2 Probabilistic Prediction Using a Finite Dataset

The main obstacle in applying Stone's theorem is that the convergence it asserts is not uniform. The situation that we typically encounter in practice is that we are given a set of examples and a new object, and we would like to estimate the probability that the label of the new object is 1. It is well understood that in this situation the applicability of Stone's theorem is very limited (see, e.g., [82, Sect. 7.1]). In Chap. 5 we give another, more direct, formalization of this observation.

We say that a dataset consisting of old examples and one new object is *diverse* if no object in it is repeated (in particular, the new object is different from all old objects). The main result of Sect. 5.3 asserts that any nontrivial (not empty and not containing 0 and 1) valid prediction interval for the conditional probability given the new object that the new label is 1 is inadmissible if the dataset is diverse and randomness is the only assumption.

The assumption that the dataset is diverse is related to the assumption of a high-dimensional environment. If the objects are, for example, complex images, we will not expect precise repetitions among them.

1.5.3 Venn Prediction

The results of Chap. 5 show that it is impossible to estimate the true conditional probabilities under the conditions stated; that chapter also contains a result (Theorem 5.5) that it is impossible to find conditional probabilities that are as good (in the sense of the algorithmic theory of randomness) as the true probabilities. If, however, we are prepared to settle for less and only want probabilities that are "well calibrated" (in other words, have a frequentist justification), a modification of conformal predictors which we call Venn predictors will achieve this goal, in a strong non-asymptotic sense. This is the subject of Chap. 6.

1.5.4 Conformal Predictive Distributions

Venn predictors only work in the case of classification. It turns out that in the case of regression conformal prediction is capable of outputting predictive distribution functions, which we call conformal predictive distributions. This is the topic of Chap. 7. It is interesting that while our approach to probabilistic prediction in the case of classification requires a nontrivial modification of conformal prediction, no such modifications are needed for regression. An application of conformal predictive distributions discussed in Chap. 7 is to decision making: practically meaningful loss functions can be combined with conformal predictive distributions to arrive at optimal decisions.

1.6 Beyond Randomness

In this book we also consider testing the assumption of randomness and alternatives to this assumption. The most radical alternative is introduced in Chaps. 11 and 12 under the name of "online compression modelling".

1.6.1 Testing Randomness

This is the topic of Part III. We start in Chap. 8 by adapting the mathematical apparatus developed in the previous chapters to testing the assumption of randomness. The usual statistical approach to testing (sometimes called the "Neyman–Pearson–Wald" theory) is essentially offline: in the original Neyman–Pearson approach (see, e.g., [222]), the sample size is chosen a priori, and in Wald's [419] sequential analysis, the sample size is data-dependent but still at some point a categorical decision on whether the null hypothesis is rejected or not is taken (with probability one). Our approach is online: we constantly update the strength of evidence against the null hypothesis of randomness. Finding evidence against the null hypothesis involves gambling against it, and the strength of evidence equals the gambler's current capital. For further details and the history of this approach to testing, see [316, 317, 320]. The main mathematical finding of Sect. 8.1 is that there exists a wide family of "exchangeability martingales", which can be successfully applied to detecting lack of randomness. In the rest of Chap. 8 we discuss different strategies for deciding when a machine-learning algorithm needs to be retrained because of the change in the distribution of the data.

In Chap. 9 we explore the efficiency of our approach to testing randomness, which we call *conformal testing*. (It turns out surprisingly efficient, although we only consider the binary case.) And in Chap. 10 we adapt the ideas of conformal testing to protecting machine-learning algorithms against changes in the distribution of the data.

1.6.2 Online Compression Models

As we will see in Chap. 11, the idea of conformal prediction generalizes from learning under randomness, where examples are IID, to "online compression models". In an online compression model, it is assumed that the data can be summarized in a way that can be updated as new examples come in, and the only known probabilities are backward probabilities—probabilities, given a summary, for the previous summary and the last example summarized.

Online compression models derive from the work of Andrei Kolmogorov. They open a new direction for broadening the applicability of machine-learning methods, giving a new meaning to the familiar idea that learning can be understood as information compression.

In Chap. 11 we consider in detail two important online compression models (exchangeability and Gaussian) and their variants. In Chap. 12 we extend the idea of Venn prediction to online compression modelling and introduce another interesting model of this kind, the hypergraphical model, which can be used in causal inference.

1.7 Context

Each chapter of this book ends with a section entitled "Context". These sections are set in a small font and may use mathematical notions and results not introduced elsewhere in the book. Almost all of them are preceded by a section (also set in a small font) containing longer proofs.

Turing suggested the idea of machine learning in his paper published in *Mind* as an approach to solving his famous "imitation game" [358, Sect. 1].

The terms "transduction" and "transductive inference" were introduced in machine learning by Vapnik [365, Sect. 6.1] in 1995. His setting of transduction, without using this terminology, was introduced in his 1971 booklet [363] and then repeated in [370] and [364]. See [366, Comments and bibliographical remarks, Chap. 8] and [367, Chap. 3] for further historical information. For a very readable review of Johnson's eduction, see Dale [68].

An alternative term for our "prediction rule" that is popular in machine learning is "prediction model". We do not use "prediction model" since the word "model" is already overloaded in this book.

An empirical study of various bounds on prediction accuracy is reported in [211]. It found the hold-out estimate to be a top performer.

Part I
Set Prediction

Chapter 2
Conformal Prediction: General Case and Regression



Abstract In this chapter we formally introduce conformal predictors. After giving the necessary definitions, we will prove that when a conformal predictor is used in the online mode, its output is valid, not only in the asymptotic sense that the sets it predicts for any fixed confidence level $1 - \epsilon$ will be wrong with frequency at most ϵ (approaching ϵ in the case of smoothed conformal predictors) in the long run, but also in a much more precise sense: the error probability of a smoothed conformal predictor is ϵ at every trial and errors happen independently at different trials. In Sect. 2.6 we will see that conformal prediction is indispensable for achieving this kind of validity. The basic procedure of conformal prediction might look computationally inefficient when the label set is large, but in Sect. 2.3 we show that in the case of, e.g., least squares regression (where the label space $\mathbb{R}$ is uncountable) there are ways of making conformal predictors much more efficient. In addition to validity, we also discuss the efficiency of conformal predictors.

Keywords Conformal prediction · Underlying algorithm · Conformalized ridge regression · Conformalized nearest neighbours regression

2.1 Confidence Predictors

The conformal predictors we define in this chapter are confidence predictors—they make a range of successively more specific predictions with successively less confidence. In this section we define precisely what we mean by a confidence predictor and its validity.

2.1.1 Assumptions

We assume that Reality outputs successive pairs

$$(x_1, y_1), (x_2, y_2), \dots , \tag{2.1}$$

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V. Vovk et al., *Algorithmic Learning in a Random World*,
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called *examples*. Each example (x_i, y_i) consists of an *object* x_i and its *label* y_i. The objects are elements of a measurable space $\mathbf{X}$ called the *object space*, and the labels are elements of a measurable space $\mathbf{Y}$ called the *label space*. We assume that $\mathbf{X}$ is nonempty and that $\mathbf{Y}$ contains at least two essentially different elements.[1] When we need a more compact notation, we write z_i for (x_i, y_i). We set $\mathbf{Z} := \mathbf{X} \times \mathbf{Y}$ and call $\mathbf{Z}$ the *example space*. Thus the infinite data sequence (2.1) is an element of the measurable space $\mathbf{Z}^\infty$.

When we say that the objects are *absent*, we mean that $|\mathbf{X}| = 1$. In this case x_i do not carry any information and do not need to be mentioned; we will then identify $\mathbf{Y}$ and $\mathbf{Z}$ and refer to the examples as *observations*. (The distinction between examples and observations will not always be clear-cut.)

Our standard assumption is that Reality chooses the examples independently from some probability distribution Q on $\mathbf{Z}$—i.e., that the infinite sequence $z_1, z_2, \dots$ is drawn from the *power probability measure* (or *power distribution*) Q^∞ in $\mathbf{Z}^\infty$. Most of the results of this book hold under this *randomness assumption*, or *randomness model*, but usually we need only the slightly weaker assumption that the infinite data sequence (2.1) is drawn from a distribution P on $\mathbf{Z}^\infty$ that is *exchangeable*. The statement that P is exchangeable means that for every positive integer n, every permutation π of $\{1, \dots, n\}$, and every measurable set $E \subseteq \mathbf{Z}^n$,

$$P\left\{(z_1, z_2, \dots) \in \mathbf{Z}^\infty : (z_1, \dots, z_n) \in E\right\} = P\left\{(z_1, z_2, \dots) \in \mathbf{Z}^\infty : (z_{\pi(1)}, \dots, z_{\pi(n)}) \in E\right\} . \tag{2.2}$$

(In other words, the distribution of the examples does not depend on their order.) Every power distribution is exchangeable, and under a natural regularity condition ($\mathbf{Z}$ is a standard Borel space), any exchangeable distribution on $\mathbf{Z}^\infty$ is a mixture of power distributions; for details, see Sect. A.5. In our mathematical results, we usually use the randomness assumption or the exchangeability assumption depending on which one leads to a stronger statement.

2.1.2 Simple Predictors and Confidence Predictors

We assume that at the nth trial Reality first announces the object x_n and only later announces the label y_n. If we simply want to predict y_n, then we need a function

$$D : \mathbf{Z}^* \times \mathbf{X} \to \mathbf{Y} . \tag{2.3}$$

[1] Formally, the σ-algebra on $\mathbf{Y}$ is assumed to be different from $\{\emptyset, \mathbf{Y}\}$. It is convenient to assume that for each pair of distinct elements of $\mathbf{Y}$ there is a measurable set containing only one of them; we will do this without loss of generality, and then our assumption about $\mathbf{Y}$ is that $|\mathbf{Y}| > 1$.

We call such a function a *simple predictor*, always assuming it is measurable. For any sequence of old examples, say $x_1, y_1, \ldots, x_{n-1}, y_{n-1} \in \mathbf{Z}^*$, and any new object, say $x_n \in \mathbf{X}$, it gives $D(x_1, y_1, \ldots, x_{n-1}, y_{n-1}, x_n) \in \mathbf{Y}$ as its prediction for the new label y_n.

As we explained in Sect. 1.4, however, we have a more complicated notion of prediction. Instead of merely choosing a single element of $\mathbf{Y}$ as our prediction for y_n, we want to give a range of more or less precise predictions, each labelled with a degree of confidence. We want to give subsets of $\mathbf{Y}$ large enough that we can be confident that y_n will fall in them, while also giving smaller subsets in which we are less confident. An algorithm that predicts in this sense requires an additional input $\epsilon \in (0, 1)$, which we call the *significance level*; the complementary value $1 - \epsilon$ is called the *confidence level*. Given all these inputs, say $x_1, y_1, \ldots, x_{n-1}, y_{n-1}, x_n, \epsilon$, an algorithm of the type that interests us, say Γ, outputs a subset

$$\Gamma_n^\epsilon = \Gamma^\epsilon (x_1, y_1, \ldots, x_{n-1}, y_{n-1}, x_n)$$

of $\mathbf{Y}$. (We position ϵ as a superscript instead of placing it with the other arguments.) We require this subset to shrink as ϵ is increased: Γ must satisfy

$$\Gamma^{\epsilon_1} (x_1, y_1, \ldots, x_{n-1}, y_{n-1}, x_n) \subseteq \Gamma^{\epsilon_2} (x_1, y_1, \ldots, x_{n-1}, y_{n-1}, x_n) \tag{2.4}$$

whenever $\epsilon_1 \geq \epsilon_2$. Intuitively, once we observe the *incomplete data sequence*

$$x_1, y_1, \ldots, x_{n-1}, y_{n-1}, x_n \tag{2.5}$$

and choose the significance level ϵ, Γ predicts that

$$y_n \in \Gamma^\epsilon (x_1, y_1, \ldots, x_{n-1}, y_{n-1}, x_n) \ , \tag{2.6}$$

and the smaller ϵ is the more confident the prediction is. According to condition (2.4), we are more confident in less specific predictions.

Formally, we call a measurable function

$$\Gamma : \mathbf{Z}^* \times \mathbf{X} \times (0, 1) \to 2^{\mathbf{Y}} \tag{2.7}$$

($2^{\mathbf{Y}}$ is the set of all subsets of $\mathbf{Y}$) that satisfies (2.4), for all significance levels $\epsilon_1 \geq \epsilon_2$, all positive integers n, and all incomplete data sequences (2.5), a *confidence predictor* (or *deterministic confidence predictor*). The requirement that Γ be measurable means that for each n the set of sequences $\epsilon, x_1, y_1, \ldots, x_n, y_n$ satisfying (2.6) is a measurable subset of $(0, 1) \times (\mathbf{X} \times \mathbf{Y})^n$.

2.1.3 Validity

Let us begin with an intuitive explanation of the notions of exact and conservative validity. For each significance level ϵ, we want to have confidence $1 - \epsilon$ in our prediction (2.6) about y_n. This means that the probability of the prediction being in error—the probability of the event (2.6) not happening—should be ϵ. Moreover, since we are making a whole sequence of predictions, first for y_1, then for y_2, and so on, we would like these error events to be independent. If these conditions are met no matter what exchangeable probability distribution P governs the sequence of examples, then we say that the confidence predictor Γ is "exactly valid", or, more briefly, "exact". This means that making errors with Γ is like getting heads when making independent tosses of a biased coin whose probability of heads is always ϵ. If the probabilities for errors are allowed to be even less than this, then we say that the confidence predictor Γ is "conservatively valid" or, more briefly, "conservative".

When we make independent tosses of a biased coin whose probability of heads is always ϵ, the frequency of heads will converge to ϵ with probability one—this is the strong law of large numbers. So the frequency of errors at significance level ϵ for an exactly valid confidence predictor converges to ϵ with probability one. As we will see, confidence predictors sometimes have this asymptotic property even when they are not exactly valid. So we give the property a name of its own; we call a confidence predictor that has it "asymptotically exact". Similarly, we call a confidence predictor for which the frequency of errors is asymptotically no more than ϵ (for which the upper limit of the frequency of errors is at most ϵ) with probability one "asymptotically conservative".

In order to restate these definitions in a way sufficiently precise to exclude possible misunderstandings, we now introduce a formal notation for the errors Γ makes when it processes the data sequence

$$\omega = (x_1, y_1, x_2, y_2, \dots)$$

at significance level ϵ. Whether Γ makes an error on the nth trial can be represented by a number that is 1 in the case of an error and 0 in the case of no error:

$$\begin{aligned} \operatorname{err}_n^\epsilon(\Gamma, \omega) &:= 1_{y_n \notin \Gamma^\epsilon(x_1, y_1, \dots, x_{n-1}, y_{n-1}, x_n)} \\ &:= \begin{cases} 1 & \text{if } y_n \notin \Gamma^\epsilon(x_1, y_1, \dots, x_{n-1}, y_{n-1}, x_n) \\ 0 & \text{otherwise} , \end{cases} \end{aligned} \tag{2.8}$$

and the number of errors during the first n trials is

$$\operatorname{Err}_n^\epsilon(\Gamma, \omega) := \sum_{i=1}^{n} \operatorname{err}_i^\epsilon(\Gamma, \omega) . \tag{2.9}$$

Protocol 2.1 Online confidence prediction

1: $\mathrm{Err}_0^\epsilon := 0$ for all $\epsilon \in (0, 1)$
2: **for** $n = 1, 2, \ldots$
3: Reality outputs $x_n \in \mathbf{X}$
4: Predictor outputs $\Gamma_n^\epsilon \subseteq \mathbf{Y}$ for all $\epsilon \in (0, 1)$
5: Reality outputs $y_n \in \mathbf{Y}$
6: $\mathrm{err}_n^\epsilon := 1_{y_n \notin \Gamma_n^\epsilon}$ for all $\epsilon \in (0, 1)$
7: $\mathrm{Err}_n^\epsilon := \mathrm{Err}_{n-1}^\epsilon + \mathrm{err}_n^\epsilon$ for all $\epsilon \in (0, 1)$.

The online learning protocol with the error counts err_n^ϵ and Err_n^ϵ is shown as Protocol 2.1. This is a game protocol, and we can consider strategies for both players, Reality and Predictor. We have assumed that Reality's strategy is randomized; her moves are generated from an exchangeable probability distribution P on $\mathbf{Z}^\infty$. A confidence predictor Γ is, by definition, a measurable strategy for Predictor.

If ω is drawn from an exchangeable probability distribution P, the number $\mathrm{err}_n^\epsilon(\Gamma, \omega)$ is the realized value of a random variable, which we designate $\mathrm{err}_n^\epsilon(\Gamma)$. Formally, the confidence predictor Γ is *exactly valid* if for each ϵ,

$$\mathrm{err}_1^\epsilon(\Gamma), \mathrm{err}_2^\epsilon(\Gamma), \ldots \tag{2.10}$$

is a sequence of independent Bernoulli random variables with parameter ϵ—i.e., if it is a sequence of independent random variables each of which has probability ϵ of being one and probability $1 - \epsilon$ of being zero—no matter what exchangeable distribution P we draw ω from. Unfortunately, the notion of exact validity is vacuous for confidence predictors: see Theorem 2.2 below.

The notion of conservative validity is more complex; now we only require that $\mathrm{err}_n^\epsilon(\Gamma)$ be *dominated* by a sequence of independent Bernoulli random variables with parameter ϵ. Let $\mathbf{U}$ be the uniform probability measure on $[0, 1]$. For each probability measure P on $\mathbf{Z}^\infty$, the probability space

$$(\mathbf{Z}^\infty \times [0, 1], P \times \mathbf{U}) \tag{2.11}$$

is the product measurable space $\mathbf{Z}^\infty \times [0, 1]$ equipped with the product probability measure $P \times \mathbf{U}$. Formally, the confidence predictor Γ is *conservatively valid* if, for any exchangeable probability distribution P on $\mathbf{Z}^\infty$, there exists a family

$$\xi_n^{(\epsilon)}, \quad \epsilon \in (0, 1), \quad n = 1, 2, \ldots , \tag{2.12}$$

of $\{0, 1\}$-valued random variables on the product probability space $(\mathbf{Z}^\infty \times [0, 1], P \times \mathbf{U})$ such that:

- for all n and ϵ, $\xi_n^{(\epsilon)}$ depends only on the first n examples;
- for any fixed ϵ, $\xi_1^{(\epsilon)}, \xi_2^{(\epsilon)}, \ldots$ is a sequence of independent Bernoulli random variables with parameter ϵ on the probability space $(\mathbf{Z}^\infty \times [0, 1], P \times \mathbf{U})$;
- for all n and ϵ, $\mathrm{err}_n^\epsilon(\Gamma) \le \xi_n^{(\epsilon)}$.

Intuitively, the auxiliary probability space $([0,1], \mathbf{U})$ represents extra randomization; the random variables $\mathrm{err}_n^\epsilon(\Gamma)$ are extended to the product probability space $\mathbf{Z}^\infty \times \mathbf{U}$ in a natural way, by ignoring the extra randomization.

To conclude, we define asymptotic validity. The confidence predictor Γ is *asymptotically exact* if, for any exchangeable probability distribution P on $\mathbf{Z}^\infty$ generating examples $z_1, z_2, \ldots$ and any significance level ϵ,

$$\lim_{n\to\infty} \frac{\mathrm{Err}_n^\epsilon(\Gamma)}{n} = \epsilon \tag{2.13}$$

with probability one. It is *asymptotically conservative* if, for any exchangeable probability distribution P on $\mathbf{Z}^\infty$ and any significance level ϵ,

$$\limsup_{n\to\infty} \frac{\mathrm{Err}_n^\epsilon(\Gamma)}{n} \leq \epsilon \tag{2.14}$$

with probability one.

Proposition 2.1 *An exact confidence predictor is asymptotically exact. A conservative confidence predictor is asymptotically conservative.*

This proposition is an immediate consequence of the law of large numbers.

2.1.4 Randomized Confidence Predictors

We will also be interested in randomized confidence predictors, which depend, additionally, on elements of an auxiliary probability space. The main advantage of randomization in this context is that, as we will see, there are many randomized confidence predictors that are exactly valid. Formally, we define a *randomized confidence predictor* to be a measurable function

$$\Gamma : (\mathbf{X} \times [0,1] \times \mathbf{Y})^* \times (\mathbf{X} \times [0,1]) \times (0,1) \to 2^{\mathbf{Y}} \tag{2.15}$$

which, for all significance levels $\epsilon_1 \geq \epsilon_2$, all positive integer n, and all incomplete data sequences

$$x_1, \tau_1, y_1, \ldots, x_{n-1}, \tau_{n-1}, y_{n-1}, x_n, \tau_n \ , \tag{2.16}$$

where $x_i \in \mathbf{X}$, $\tau_i \in [0,1]$, and $y_i \in \mathbf{Y}$ for all i, satisfies

$$\begin{aligned} &\Gamma^{\epsilon_1}(x_1, \tau_1, y_1, \ldots, x_{n-1}, \tau_{n-1}, y_{n-1}, x_n, \tau_n) \\ &\qquad \subseteq \Gamma^{\epsilon_2}(x_1, \tau_1, y_1, \ldots, x_{n-1}, \tau_{n-1}, y_{n-1}, x_n, \tau_n) \ . \end{aligned} \tag{2.17}$$

We will always assume that $\tau_1, \tau_2, \ldots$ are random variables that are independent (between themselves and of anything else) and distributed uniformly on $[0, 1]$; this is what one expects to get from a random number generator. (Usually there is no dependence of Γ on $\tau_1, \ldots, \tau_{n-1}$ in its argument (2.16).)

We define $\mathrm{err}^\epsilon_n(\Gamma, \omega)$ and $\mathrm{Err}^\epsilon_n(\Gamma, \omega)$ as before, only now they also depend on the τ's. In other words, $\mathrm{err}^\epsilon_n(\Gamma, \omega)$ is defined by (2.8) with x_i now being *extended objects* $x_i \in \mathbf{X} \times [0, 1]$; $\mathrm{Err}^\epsilon_n(\Gamma, \omega)$ is defined, as before, by (2.9).

In general, many definitions for randomized confidence predictors are special cases of the corresponding definitions for deterministic confidence predictors: the latter should just be applied to the *extended object space* $\mathbf{X} \times [0, 1]$ (whose elements consist of both the object and the random number to be used at a given trial).

If Γ is a randomized confidence predictor, P is an exchangeable distribution on $\mathbf{Z}^\infty$, and $n \in \mathbb{N}$,

$$\mathrm{err}^\epsilon_n(\Gamma, (x_1, \tau_1, y_1, x_2, \tau_2, y_2, \ldots)) \text{ and } \mathrm{Err}^\epsilon_n(\Gamma, (x_1, \tau_1, y_1, x_2, \tau_2, y_2, \ldots)) \,,$$

where $(x_1, y_1), (x_2, y_2), \ldots$ are drawn from P and $\tau_1, \tau_2, \ldots$ are drawn independently from $\mathbf{U}^\infty$, the uniform distribution on $[0, 1]^\infty$, are random variables, which will be denoted $\mathrm{err}^\epsilon_n(\Gamma)$ and $\mathrm{Err}^\epsilon_n(\Gamma)$, respectively. We say that Γ is *exactly valid* if for each $\epsilon \in (0, 1)$, (2.10) is a sequence of independent Bernoulli random variables with parameter ϵ.

We are not really interested in the notion of conservative validity for randomized confidence predictors.

2.1.5 Confidence Predictors Over a Finite Horizon

A trivial modification of online prediction is were Reality generates only N examples $z_1, \ldots, z_N$. We have the same protocol, Protocol 2.1, except that n now goes from 1 to N. We will refer to it as the *online protocol with horizon* N.

This modification, however, is important, because it allows us to relax the assumption about the data-generating mechanism and still have valid confidence predictors. We say that a probability measure P on $\mathbf{Z}^N$ is *exchangeable* if we have the following simplified version of (2.2): for every permutation π of $\{1, \ldots, N\}$ and every measurable $E \subseteq \mathbf{Z}^N$,

$$P(E) = P\left\{(z_1, \ldots, z_N) \in \mathbf{Z}^N : (z_{\pi(1)}, \ldots, z_{\pi(N)}) \in E\right\} \,.$$

The definitions of conservative and exact validity directly carry over to the online protocol with horizon N and exchangeable data-generating measures on $\mathbf{Z}^N$. Achieving validity in this sense (e.g., by the method of conformal prediction, as

discussed in the next section) is valuable since the assumption that $z_1, \dots, z_N$ are exchangeable is much weaker than the assumption that the sequence can be extended to a longer exchangeable sequence $z_1, \dots, z_{N+1}$, let alone the assumption that it can be extended to an infinite exchangeable sequence $z_1, z_2, \dots$. (See Sect. 2.9.1 below.)

The requirement of exchangeability of examples $z_1, \dots, z_N$ in a benchmark dataset (such as the USPS dataset) is often violated (see, e.g., Sect. 8.1.2), but it is easy to enforce by randomly permuting it. This way we will enforce the exchangeability of the examples, whereas there is no easy way to enforce their randomness.

2.1.6 One-Off and Offline Confidence Predictors

Let us start from a terminological remark. In the offline mode of prediction we are given two collections of examples, which we usually refer to as the *training set* and *test set* (as we already did in Chap. 1), in order to use familiar terms. In the one-off mode, the test set consists of only one example. Formally the training and test sets, however, are not sets but sequences. The order of examples in them often does not matter, but even in such cases, they are still not sets since they may contain the same element more than once (and so they are bags in the terminology that will be introduced in Sect. 2.2.1).

In the one-off mode we are given a training set $z_1, \dots, z_l$ of examples $z_i = (x_i, y_i)$ and a *test object* $x_{l+1} \in \mathbf{X}$, and the problem is to predict the label y_{l+1} of x. A *one-off confidence predictor* is a measurable function

$$\Gamma : \mathbf{Z}^l \times \mathbf{X} \times (0, 1) \to 2^{\mathbf{Y}}$$

(cf. (2.7)) that always satisfies (2.4), where $n := l + 1$. Similarly, a *one-off randomized confidence predictor* is a measurable function

$$\Gamma : (\mathbf{X} \times [0, 1] \times \mathbf{Y})^l \times (\mathbf{X} \times [0, 1]) \times (0, 1) \to 2^{\mathbf{Y}}$$

(cf. (2.15)) that always satisfies (2.17) with $n := l + 1$.

Let us say that a one-off confidence predictor is *conservatively valid under randomness* if its probability of error at any significance level $\epsilon \in (0, 1)$ is at most ϵ under any power distribution Q^{l+1} on the examples. Being *conservatively valid under exchangeability* is defined in the same way except for replacing the power distributions with the exchangeable distributions on $\mathbf{Z}^{l+1}$. Finally, in being *conservatively valid under infinite exchangeability* the assumption of exchangeability of $z_1, \dots, z_{l+1}$ is replaced by the assumption that those examples are the first $l + 1$ in an infinite exchangeable sequence $z_1, z_2, \dots$. All these notions of validity are weak in that they do not involve any claim of independence of errors (similar to what we required for conservative or exact validity of online predictors).

The task of designing valid one-off confidence predictors can be set under various assumptions, such as randomness, infinite exchangeability (where the available examples are the first $l + 1$ of an infinite exchangeable sequence), and the exchangeability of only the available examples $z_1, \ldots, z_l, (x_{l+1}, y_{l+1})$. In the next section we will see that even the last version is feasible.

For one-off confidence predictors, both randomized and deterministic, we have the useful notion of *exact validity under randomness*, i.e., their probability of error at any significance level $\epsilon \in (0, 1)$ being exactly ϵ under any power distribution Q^{l+1} on the examples. In the same way we define *exact validity under infinite exchangeability* and *exact validity under exchangeability*. These notions, however, are only useful for randomized predictors.

Theorem 2.2 *No one-off confidence predictor is exactly valid under randomness.*

See Sect. 2.8.1 for a proof.

One-off confidence predictors can be used in the offline setting as well, where, along with the training set $z_1, \ldots, z_l$, we also have a test set $z_{l+1}, \ldots, z_{l+k}$, where $z_i = (x_i, y_i)$. For each test object x_i, the prediction set for y_i is $\Gamma^\epsilon(z_1, \ldots, z_l, x_i)$.

2.2 Conformal Predictors

We start by defining the concept of a nonconformity measure. Intuitively, this is a way of measuring how strange an example is in a bag of examples. There are many different nonconformity measures, and each one defines a conformal predictor and a smoothed conformal predictor.

2.2.1 Bags

The order in which examples appear often does not make any difference, and in order to formalize this point, we need the concept of a bag (also called a multiset). A *bag* of size $n \in \mathbb{N}$ is a collection of n elements some of which may be identical; a bag resembles a set in that the order of its elements is irrelevant, but it differs from a set in that repetition is allowed. To identify a bag, we must say what elements it contains and how many times each of these elements is repeated. We write $\{\!|z_1, \ldots, z_n|\!\}$ for the bag consisting of elements $z_1, \ldots, z_n$, some of which may be identical with each other. The empty bag $\{\!|\,|\!\}$ contains no elements. We are only interested in finite bags.

We write $Z^{(n)}$ for the set of all bags of size n of elements of a measurable space Z. The set $Z^{(n)}$ is itself a measurable space. It can be defined formally as the power space Z^n with a non-standard σ-algebra, consisting of measurable subsets of Z^n that contain all permutations of their elements. We write $Z^{(*)}$ for the set of all bags of elements of Z (the union of all the $Z^{(n)}$).

Operations such as union ($\cup$), intersection ($\cap$), and difference ($\setminus$) are defined for bags in a natural way. For example, if $A, B \in Z^{(*)}$ are two bags, the number of times their union $A \cup B$ contains any $z \in Z$ is the number of times A contains z plus the number of times B contains z.

2.2.2 Nonconformity and Conformity

As we have already said, a nonconformity measure is a way of scoring how strange an example is as an element of a bag. Formally, a *nonconformity measure* is a measurable mapping

$$A : \mathbf{Z}^{(*)} \times \mathbf{Z} \to \overline{\mathbb{R}} ; \tag{2.18}$$

to each possible bag of examples and each possible example, in or outside the bag, A assigns a numerical score indicating how different the new example is from the old ones. In our definitions we will ignore the values $A(\sigma, z)$ for examples $z \notin \sigma$.

It is easy to invent nonconformity measures, especially when we already have methods for prediction at hand:

- If the examples are merely numbers ($\mathbf{Z} = \mathbb{R}$), then we might define the nonconformity of an example as the absolute value of its difference from the average of all examples in the bag. Alternatively, we could use instead the absolute value of its difference from the median of all examples.
- In a regression problem where examples are pairs of numbers, say $z_i = (x_i, y_i)$, we might define the nonconformity of an example (x, y) as the absolute value of the difference between y and the predicted value $\hat{y}$ calculated from x and all examples in the bag.

We will discuss the latter way of detecting nonconformity further at the end of this section. But whether a particular function on $\mathbf{Z}^{(*)} \times \mathbf{Z}$ is an appropriate way of measuring nonconformity will always be open to discussion, and we do not need to enter into this discussion at this point. In our general theory, we will call *any* measurable function on $\mathbf{Z}^{(*)} \times \mathbf{Z}$ taking values in the extended real line a nonconformity measure.

Given a nonconformity measure A, a sequence $z_1, \dots, z_n$ of examples, and an example z, we can score how different z is from the bag $\{z_1, \dots, z_n\}$; namely, $A(\{z_1, \dots, z_n\}, z)$ is called the *nonconformity score* for z. We will sometimes refer to $\{z_1, \dots, z_n\}$ as the *comparison bag* in this context.

Of course, instead of looking at functions that we feel measure nonconformity, we could look at functions that we feel measure conformity. We call such a function, say B, a conformity measure, and we can use it to define *conformity scores* $B(\{z_1, \dots, z_n\}, z)$. Formally, a *conformity measure* is a measurable function of the type $\mathbf{Z}^{(*)} \times \mathbf{Z} \to \overline{\mathbb{R}}$ (so there is no difference between conformity measures and nonconformity measures as mathematical objects). If we begin in this way, then

nonconformity appears as a derivative idea. Given a conformity measure B we can define a nonconformity measure A using any strictly decreasing transformation, say $A := -B$ or perhaps (if B takes only positive values) $A := 1/B$. Given our goal, prediction, beginning with conformity might seem the more natural approach. As we will explain shortly, our strategy for prediction is to predict that a new label will be among the labels that best make a new example conform with old examples, and it is more natural to emphasize the labels that we include in the prediction (the most conforming ones) rather than the labels that we exclude (the most nonconforming ones). But in practice, it is often more natural to begin with nonconformity measures. For example, when we compare a new example with an average of old examples, we will usually first define a distance between the two rather than devise a way to measure their closeness. For this reason, we often (but far from always) emphasize nonconformity rather than conformity. Doing so is consistent with tradition in mathematical statistics, where test statistics are usually defined so as to measure discrepancy rather than agreement.

2.2.3 p-Values

Given a nonconformity measure A and a bag $\wr z_1, \ldots, z_n \int$, we can compute the nonconformity score

$$\alpha_i := A\left(\wr z_1, \ldots, z_n \int, z_i\right) \tag{2.19}$$

for each example z_i in the bag. Because a nonconformity measure A may be scaled however we like, the numerical value of α_i does not, by itself, tell us how unusual A finds z_i to be. For that, we need a comparison of α_i to the other α_j. A convenient way of making this comparison is to compute the fraction

$$\frac{\left|\{j = 1, \ldots, n : \alpha_j \geq \alpha_i\}\right|}{n} . \tag{2.20}$$

This fraction, which lies between $1/n$ and 1, is called the *p-value* for z_i. It is the fraction of the examples in the bag as nonconforming as z_i. If it is small (close to its lower bound $1/n$ for a large n), then z_i is very nonconforming (an outlier). If it is large (close to its upper bound 1), then z_i is very conforming.

If we begin with a conformity measure rather than a nonconformity measure, then we can define the p-value for z_i by

$$\frac{\left|\{j = 1, \ldots, n : \beta_j \leq \beta_i\}\right|}{n} ,$$

where the β_j are the conformity scores. This gives the same result as we would obtain from (2.20) using a nonconformity measure A obtained from B by means of a strictly decreasing transformation.

2.2.4 Definition of Conformal Predictors

Every nonconformity measure determines a confidence predictor. Given a new object x_n and a level of significance, this predictor provides a prediction set that should contain the object's label y_n. We obtain the set by supposing that y_n will have a value that makes (x_n, y_n) conform with the previous examples. The level of significance determines the amount of conformity (as measured by the p-value) that we require.

Formally, the *conformal predictor determined by a nonconformity measure* A is the confidence predictor Γ obtained by setting

$$\Gamma^{\epsilon}(x_1, y_1, \ldots, x_{n-1}, y_{n-1}, x_n) \tag{2.21}$$

equal to the set of all labels $y \in \mathbf{Y}$ such that

$$\frac{|\{i = 1, \ldots, n : \alpha_i \geq \alpha_n\}|}{n} > \epsilon\ , \tag{2.22}$$

where

$$\begin{aligned} \alpha_i &:= A\big(\sigma, (x_i, y_i)\big), \quad i = 1, \ldots, n-1, \\ \alpha_n &:= A\big(\sigma, (x_n, y)\big), \\ \sigma &:= \{(x_1, y_1), \ldots, (x_{n-1}, y_{n-1}), (x_n, y)\}\ . \end{aligned} \tag{2.23}$$

We will sometimes refer to σ defined in this way as the *augmented training set*. In general, a *conformal predictor* is a conformal predictor determined by some nonconformity measure.

The left-hand side of (2.22) is the p-value of (x_n, y) in the bag consisting of it and the old examples (cf. (2.20)). So our prediction with significance level ϵ (or confidence level $1 - \epsilon$) is that the value of y_n will make (x_n, y_n) have a p-value greater than ϵ when it is bagged with the old examples. We are 98% confident, for example, that we will get a value for y_n that gives (x_n, y_n) a p-value greater than 0.02. In other words, we are 98% confident that

$$\frac{\text{number of examples that conform worse or the same as } (x_n, y_n)}{n}$$

will exceed 0.02.

If A is a conformity measure, the conformal predictor determined by A is defined by (2.22) with "$\geq$" replaced by "$\leq$".

2.2.5 Validity

Proposition 2.3 *All conformal predictors are conservative.*

It follows by Proposition 2.1 that a conformal predictor is asymptotically conservative. Of course, more can be said. Using the law of the iterated logarithm instead of the law of large numbers, we can strengthen (2.14) to

$$\limsup_{n\to\infty} \frac{\mathrm{Err}_n^\epsilon(\Gamma) - n\epsilon}{\sqrt{2\epsilon(1-\epsilon)n \ln\ln n}} \leq 1 \ .$$

We will also state two finite-sample implications of Proposition 2.3: Hoeffding's inequality (see Sect. A.6.3) implies that, for any positive integer N and any constant $\delta > 0$,

$$P\left\{\omega : \mathrm{Err}_N^\epsilon(\Gamma, \omega) \geq N(\epsilon + \delta)\right\} \leq \mathrm{e}^{-2N\delta^2} \ ;$$

the central limit theorem implies that, for any constant c,

$$\limsup_{N\to\infty} P\left\{\omega : \mathrm{Err}_N^\epsilon(\Gamma, \omega) \geq N\epsilon + c\sqrt{N}\right\} \leq \Phi\left(-c/\sqrt{\epsilon(1-\epsilon)}\right) \ ,$$

where Φ is the standard normal distribution function. For a graphical illustration of asymptotic conservativeness, see Fig. 1.6.

2.2.6 Smoothed Conformal Predictors

In this section we introduce a modification of conformal predictors which will allow us to simplify and strengthen Proposition 2.3. The *smoothed conformal predictor determined by the nonconformity measure A* is the following randomized confidence predictor Γ: the set

$$\Gamma^\epsilon\left(x_1, \tau_1, y_1, \ldots, x_{n-1}, \tau_{n-1}, y_{n-1}, x_n, \tau_n\right)$$

consists of the $y \in \mathbf{Y}$ satisfying

$$\frac{|\{i = 1, \ldots, n : \alpha_i > \alpha_n\}| + \tau_n \, |\{i = 1, \ldots, n : \alpha_i = \alpha_n\}|}{n} > \epsilon \ , \tag{2.24}$$

where α_i are defined, as before, by (2.23). The left-hand side of (2.24) is called the *smoothed p-value*.

The main difference of (2.24) from (2.22) is that in the former we treat the borderline cases $\alpha_i = \alpha_n$ more carefully. Instead of increasing the p-value by $1/n$ for each $\alpha_i = \alpha_n$, we increase it by a random amount between 0 and $1/n$.

When n is not too small, it is typical for almost all $\alpha_1, \dots, \alpha_n$ to be different, and then there is very little difference between conformal predictors and smoothed conformal predictors.

Proposition 2.4 *Any smoothed conformal predictor is exactly valid.*

This proposition will be proved in Chap. 11 (as a special case of Theorem 11.1). It immediately implies Proposition 2.3: if a smoothed conformal predictor Γ and a conformal predictor $\bar{\Gamma}$ are constructed from the same nonconformity measure, the latter's errors $\overline{\mathrm{err}}_n^\epsilon$ never exceed the former's errors err_n^ϵ, $\overline{\mathrm{err}}_n^\epsilon \le \mathrm{err}_n^\epsilon$; besides, the random numbers τ_n in the definition of conformal predictors can be merged into one random number

$$\sum_{n=1}^{\infty} \tau_n 2^{-n} \sim \mathbf{U} \tag{2.25}$$

required in the definition of conservative validity. It also implies

Corollary 2.5 *Every smoothed conformal predictor is asymptotically exact.*

2.2.7 Finite-Horizon Conformal Prediction

The definition of conformal predictors and smoothed conformal predictors also works for the online protocol with horizon N, discussed in Sect. 2.1.5; we just apply the recipe (2.21)–(2.23) (with (2.24) in place of (2.22) for smoothed conformal predictors) for $n = 1, \dots, N$.

Propositions 2.3 and 2.4 continue to hold in the online protocol with horizon N under the assumption, discussed in Sect. 2.1.5, that $z_1, \dots, z_N$ are exchangeable (rather than being the first N examples of an infinite exchangeable sequence). When talking about conservative or exact validity in the online protocol with horizon N, we will mean the definitions given before with n ranging over $1, \dots, N$ and with $z_1, \dots, z_N$ assumed exchangeable.

Proposition 2.6 *In the online protocol with a finite horizon, any conformal predictor is conservatively valid, and any smoothed conformal predictor is exactly valid.*

Proof This is a special case of Theorem 11.2 in Chap. 11.

2.2.8 One-Off and Offline Conformal Predictors

The notion of conformal predictor simplifies in the one-off and offline modes, but the price to pay is a weaker notion of validity. In the one-off mode, given a training set $z_1, \dots, z_l$, $z_i = (x_i, y_i)$, we are predicting the label of a test object $x \in \mathbf{X}$. The *one-off conformal predictor* outputs the prediction set

$$\Gamma^\epsilon(x_1, y_1, \dots, x_l, y_l, x) := \left\{ y \in \mathbf{Y} : \frac{\left|\{j = 1, \dots, l+1 : \alpha_j \geq \alpha_{l+1}\}\right|}{l+1} > \epsilon \right\} , \tag{2.26}$$

where the nonconformity scores $\alpha_1, \dots, \alpha_{l+1}$ are computed from a nonconformity measure A:

$$\begin{aligned} \alpha_j &:= A\left(\langle z_1, \dots, z_l, (x, y)\rangle, z_j\right) , \quad j = 1, \dots, l , \\ \alpha_{l+1} &:= A\left(\langle z_1, \dots, z_l, (x, y)\rangle, (x, y)\right) . \end{aligned} \tag{2.27}$$

It is a one-off confidence predictor. We will refer to $n := l + 1$ as its *size*.

According to Proposition 2.4, one-off conformal predictors are conservatively valid under infinite exchangeability and, therefore, under randomness. They are even conservatively valid under exchangeability. As we pointed out in Sect. 2.1.6, our notions of validity in the one-off mode do not involve any claim of independence of errors. An advantage of these weaker notions of validity is that they continue to hold for the following generalized notion of conformal predictor.

The definition of nonconformity scores α_i requires that they should not depend on the order of the examples in the comparison bag. Let us now drop this requirement. A *nonconformity statistic* is defined to be a measurable mapping $S : \mathbf{Z}^* \to \mathbb{R}$; this is a standard notion of a test statistic in the context of permutation tests. Now we can modify (2.26) and (2.27) as

$$\Gamma^\epsilon(z_1, \dots, z_l, x) := \left\{ y \in \mathbf{Y} : \frac{\left|\{\pi : S(z_{\pi(1)}, \dots, z_{\pi(l)}, z_{\pi(l+1)}) \geq S(z_1, \dots, z_l, z_{l+1})\}\right|}{(l+1)!} > \epsilon \right\} , \tag{2.28}$$

where $z_{l+1} := (x, y)$ and π ranges over all $(l+1)!$ permutations of the set $\{1, \dots, l+1\}$. Notice that (2.26) is a special case of (2.28) corresponding to nonconformity statistics that do not depend on the ordering of their arguments apart from the last one:

$$S(z_1, \dots, z_n) := A(\langle z_1, \dots, z_n\rangle, z_n)$$

for some nonconformity measure A. We will say that (2.28) is a *weak conformal predictor*. As in the case of conformal predictors, we will also use conformity statistics, in a similar sense to conformity measures.

In the offline mode of prediction, we are given a training set $z_1, \dots, z_l$ of examples $z_i = (x_i, y_i)$ and the problem is to predict the labels $y_i, i = l+1, \dots, l+k$, of the test examples $z_{l+1}, \dots, z_{l+k}$, given their objects. In this case we apply the one-off conformal predictor to the training set $z_1, \dots, z_l$ and the test object x_i for each $i = l + 1, \dots, l + k$ separately.

2.2.9 General Schemes for Defining Nonconformity

There are many different ways of defining nonconformity measures, but here we will only explain the most basic approaches, which in the following sections will be illustrated in the case of regression, $\mathbf{Y} = \mathbb{R}$. As already mentioned, our nonconformity measures are often based on an underlying algorithm, which is a standard prediction algorithm outputting point predictions (i.e., they are simple predictors in the terminology of Sect. 2.1.2) or probabilistic predictions (this will be relevant in Part II) without any guarantees of validity. In Sects. 2.3 and 2.4 we use ridge regression and nearest neighbours as underlying algorithms, respectively, and in Sect. 2.9.5 we give references to more interesting and sophisticated examples, such as the lasso [224]. We will give more examples in later chapters: in Sect. 3.2 for the case of classification and in Sects. 4.2–4.3 for the cases of both classification and regression. An unusual example will be given in Sect. 4.3.8, where a conformal predictor will be constructed without using an underlying simple predictor.

Conformity to a Bag

Suppose we are given a bag $\{z_1, \dots, z_n\}$ and we want to estimate the nonconformity of each example z_i as an element of the bag, as in (2.19). (It is clear that the values (2.19) are all we need to know about the nonconformity measure, and we will often define nonconformity measures by specifying the nonconformity scores (2.19).) There is a natural solution if we are given a simple predictor (2.3) whose output does not depend on the order in which the old examples are presented. The simple predictor D then defines a prediction rule $D_{\{z_1,\dots,z_n\}} : \mathbf{X} \to \mathbf{Y}$ by the formula

$$D_{\{z_1,\dots,z_n\}}(x) := D(z_1, \dots, z_n, x) \,.$$

A natural measure of nonconformity of z_i is the deviation of the predicted label

$$\hat{y}_i := D_{\{z_1,\dots,z_n\}}(x_i) \tag{2.29}$$

from the true label y_i. In this way any simple predictor, combined with a suitable measure of deviation of $\hat{y}_i$ from y_i, leads to a nonconformity measure and, therefore, to a conformal predictor.

The simplest way of measuring the deviation of $\hat{y}_i$ from y_i is to take the absolute value of their difference as α_i,

$$\alpha_i := \left| y_i - \hat{y}_i \right| . \tag{2.30}$$

Another approach is to take $\alpha_i := \left| y_i - \hat{y}_{(i)} \right|$, where $\hat{y}_{(i)}$ is the *deleted prediction*

$$\hat{y}_{(i)} := D_{\lbag z_1,\dots,z_{i-1},z_{i+1},\dots,z_n \rbag}(x_i)$$

computed by applying to x_i the prediction rule found from the dataset with the example z_i deleted. The rationale behind this deletion is that z_i, even if it is an outlier, can influence the prediction rule $D_{\lbag z_1,\dots,z_n \rbag}$ so heavily that $\hat{y}_i$ will become close to y_i, even though y_i can be very far from $\hat{y}_{(i)}$. In the rest of this subsection we will only discuss the *ordinary* predictions (2.29), but the modifications to the deleted predictions will be obvious.

More generally, the prediction rule $D_{\lbag z_1,\dots,z_n \rbag}$ may map $\mathbf{X}$ to some *prediction space* $\hat{\mathbf{Y}}$, a measurable space not necessarily coinciding with $\mathbf{Y}$. For example, the elements of $\hat{\mathbf{Y}}$ may be probability distributions on $\mathbf{Y}$, or simple predictions complemented by some measure of their accuracy (such as an estimated variance or, for multidimensional labels, covariance matrix). An *invariant predictor* is a function D that maps each bag $\lbag z_1, \dots, z_n \rbag$ of each size n to a prediction rule $D_{\lbag z_1,\dots,z_n \rbag} : \mathbf{X} \to \hat{\mathbf{Y}}$ and such that the function

$$(\lbag z_1, \dots, z_n \rbag, x) \mapsto D_{\lbag z_1,\dots,z_n \rbag}(x)$$

of the type $\mathbf{Z}^{(n)} \times \mathbf{X} \to \hat{\mathbf{Y}}$ is measurable for all n. A *discrepancy measure* is a measurable function $\Delta : \mathbf{Y} \times \hat{\mathbf{Y}} \to \overline{\mathbb{R}}$. Given an invariant predictor D and a discrepancy measure Δ, we define a function A as follows: for any $((x_1, y_1), \dots, (x_n, y_n)) \in \mathbf{Z}^*$, the nonconformity (or conformity) scores

$$\alpha_i = A\big(\lbag (x_1, y_1), \dots, (x_n, y_n) \rbag, (x_i, y_i) \big)$$

are defined by the formula

$$\alpha_i := \Delta \left(y_i, D_{\lbag (x_1,y_1),\dots,(x_n,y_n) \rbag}(x_i) \right) . \tag{2.31}$$

This way we obtain a nonconformity measure.

A simple special case of (2.31) is

$$\alpha_i := \frac{\left|y_i - \hat{y}_i\right|}{\hat{\sigma}_i} , \tag{2.32}$$

where $\hat{\sigma}_i$ is an estimate of the accuracy of $\hat{y}_i$ computed from $\lbag z_1, \dots, z_n \rbag$ and x_i. It corresponds to an invariant predictor that outputs pairs $(\hat{y}, \hat{\sigma})$ as its predictions.

Conformity to a Property

So far we talked about an important but not the only aspect of conformity, where $A(\sigma, z)$ measures how well z conforms to the bag σ. A standard feature of formalizations of this aspect, such as (2.30) or (2.32), is the presence of the operation of taking the absolute value.

Another important case is where $A(\sigma, z)$ measures how well z conforms to a property that elements of σ may satisfy to different degrees. An example is the modification

$$\alpha_i := y_i - \hat{y}_i \tag{2.33}$$

of (2.30). Defined in this way, α_i show how well z_i conforms to the property of having a large label. The modification

$$\alpha_i := \hat{y}_i - y_i \tag{2.34}$$

of (2.33) shows how well z_i conforms to the property of having a small label. The asymmetric versions (2.33) and (2.34) will often be used later in this chapter and in the rest of the book. The asymmetric versions are still special cases of the general scheme (2.31).

2.2.10 Deleted Conformity Measures

When defining a specific conformity measure A in the following chapters, we will often find it convenient to do it in terms of its *deleted version*

$$A^{\mathrm{del}}(\sigma, z) := A(\sigma \cup z, z) . \tag{2.35}$$

Of course, the conformity measure can be restored given its deleted version as

$$A(\sigma, z) = A^{\mathrm{del}}(\sigma \setminus \{z\}, z)$$

(more, precisely, its values on the relevant part of its domain, $\{(\sigma, z) : z \in \sigma\}$, can be restored).

2.3 Conformalized Ridge Regression

In this section we will implement conformal prediction based on (2.33), (2.34), or their combination concentrating on the case where the underlying simple predictor is ridge regression. In the next section we will adapt the resulting prediction algorithm to nearest neighbours regression.

Ridge regression and nearest neighbours are two of the most standard regression algorithms. In the case of regression there is an obvious difficulty in implementing the idea of conformal prediction: it appears that to form a prediction set (2.21) we need to examine each potential classification y (cf. (2.23)). We will see, however, that in some important cases there is a feasible way to compute (2.21); in particular, this is the case for ridge regression and nearest neighbours regression. (We will make little effort to optimize the computational resources required, so "feasible" essentially means "avoiding examining infinitely many cases" here. Much faster algorithms will be constructed in Sect. 4.2.)

2.3.1 Least Squares and Ridge Regression

Ridge regression and its special case, least squares, are among the most widely used regression algorithms. Least squares is the classical algorithm (going back to Gauss and Legendre), and ridge regression is its modification proposed in the 1960s. In this subsection we will describe least squares and ridge regression as simple predictors; for further details, see, e.g., [252].

Suppose $\mathbf{X} = \mathbb{R}^p$ (objects are vectors consisting of p attributes), $\mathbf{Y} = \mathbb{R}$ (we are dealing with the problem of regression), and we are given a training set $z_1, \dots, z_n$. To approximate the data, the ridge regression procedure recommends calculating the value $w \in \mathbb{R}^p$ where

$$a \|w\|^2 + \sum_{i=1}^{n} (y_i - w \cdot x_i)^2 \to \min \tag{2.36}$$

is attained; a is a nonnegative constant called the *ridge parameter*. The ridge regression prediction $\hat{y}$ for the label y of an object x is then $\hat{y} := w \cdot x$. Least squares is the special case corresponding to $a = 0$.

We can naturally represent the ridge regression procedure in a matrix form. Let Y_n be the (column-) vector

$$Y_n := (y_1, \dots, y_n)' \tag{2.37}$$

of the labels and X_n be the $n \times p$ matrix (*data matrix*) formed from the objects

$$X_n := (x_1, x_2, \dots, x_n)' \ . \tag{2.38}$$

Now we can represent the ridge regression procedure (2.36) as

$$a\|w\|^2 + \|Y_n - X_n w\|^2 \to \min , \tag{2.39}$$

or

$$Y_n' Y_n - 2w' X_n' Y_n + w'(X_n' X_n + aI_p)w \to \min ,$$

I_p being the identity $p \times p$ matrix. Taking the derivative in w we obtain

$$2(X_n' X_n + aI_p)w - 2X_n' Y_n = 0 ,$$

or

$$w = (X_n' X_n + aI_p)^{-1} X_n' Y_n . \tag{2.40}$$

Standard statistical textbooks mainly discuss the case $a = 0$ (least squares; in this case the usual assumption is that the matrix $X_n' X_n$ is invertible). It is easy to see, however, that not only least squares is a special case of ridge regression, but ridge regression can be reduced to least squares as well: the solution of (2.39) for a general $a \geq 0$ can be found as the solution to the least squares problem

$$\left\|\overline{Y} - \overline{X} w\right\|^2 \to \min ,$$

where $\overline{Y}$ is Y_n extended by adding p 0s on top and $\overline{X}$ is X_n extended by adding the $p \times p$ matrix $\sqrt{a} I_p$ on top.

2.3.2 Basic CRR

A natural conformal predictor based on ridge regression as underlying algorithm uses (2.30) as nonconformity measure, with $\hat{y}$ computed from ridge regression, and indeed this is the approach that we followed in the first edition [402, Sect. 2.3]. However, a much simpler conformal predictor can be obtained as the combination of the conformal predictor determined by (2.33) as nonconformity measure (and producing upper bounds on the true labels as its predictions) and the conformal predictor determined by (2.34) as nonconformity measure (and producing lower bounds on the true labels as its predictions). This is the strategy that we will follow.

Therefore, we start from the one-sided version of (2.31), with $\Delta(y, \hat{y}) := y - \hat{y}$, considered as nonconformity measure; we will refer to it as the *upper CRR*. The predictions will usually be prediction rays bounded above, i.e., prediction sets of the form $(-\infty, u]$ for some $u \in \mathbb{R}$. (And some applications only require this kind of predictions.)

Now the nonconformity scores α_i are simply the *residuals* $e_i := y_i - \hat{y}_i$, also known as ordinary residuals. Two slightly more sophisticated approaches will be considered in the following subsection.

From (2.40) we can see that the ridge regression prediction for an object x is

$$x'w = x'(X_n'X_n + aI_p)^{-1}X_n'Y_n \ ;$$

therefore, the predictions $\hat{y}_i$ for the objects x_i are given by

$$\hat{Y}_n := (\hat{y}_1, \ldots, \hat{y}_n)' = X_n(X_n'X_n + aI_p)^{-1}X_n'Y_n \ .$$

The matrix

$$H_n := X_n(X_n'X_n + aI_p)^{-1}X_n' \tag{2.41}$$

is called the *hat matrix* (since it transforms the y_i into the hatted form $\hat{y}_i$) and plays an important role in the standard regression theory. This matrix, as well as $I_n - H_n$, is symmetric and idempotent when $a = 0$ (remember that a symmetric matrix M is *idempotent* if $MM = M$). Therefore, the vector of nonconformity scores $(\alpha_1, \ldots, \alpha_n)'$ can be written in the form

$$Y_n - H_nY_n = (I_n - H_n)Y_n \ .$$

Now suppose that we know the incomplete data sequence (2.5), we are given a significance level ϵ, and we want to compute the prediction set output by the conformal predictor determined by the conformity scores $\alpha_i = e_i$. Let y be a possible label for x_n and

$$Y^y := (y_1, \ldots, y_{n-1}, y)' = (y_1, \ldots, y_{n-1}, 0)' + y(0, \ldots, 0, 1)'$$

be the vector of labels. The vector of nonconformity scores is $(I_n - H_n)Y^y = A + yB$, where

$$A := (I_n - H_n)(y_1, \ldots, y_{n-1}, 0)'$$
$$B := (I_n - H_n)(0, \ldots, 0, 1)' \ .$$

The components of A and B, respectively, will be denoted by $a_1, \ldots, a_n$ and $b_1, \ldots, b_n$.

If we define

$$S_i := \{y : a_i + b_i y \geq a_n + b_n y\} \ , \tag{2.42}$$

the definition of the p-values can be rewritten as

Algorithm 2.2 Upper conformalized ridge regression (simplified)

Input: a training set $(x_i, y_i) \in \mathbb{R}^p \times \mathbb{R}$, $i = 1, \dots, n-1$, and a test object $x_n \in \mathbb{R}^p$.
Set $C := I_n - X_n(X_n'X_n + aI_p)^{-1}X_n'$, X_n being defined by (2.38)
set $A = (a_1, \dots, a_n)' := C(y_1, \dots, y_{n-1}, 0)'$
set $B = (b_1, \dots, b_n)' := C(0, \dots, 0, 1)'$
for $i = 1, \dots, n-1$
    if $b_n - b_i > 0$ set $t_i := (a_i - a_n)/(b_n - b_i)$
    else set $t_i := \infty$
sort $t_1, \dots, t_{n-1}$ in the ascending order obtaining $t_{(1)} \le \dots \le t_{(n-1)}$
output $\left(-\infty, t_{(\lceil (1-\epsilon)n \rceil)}\right]$ as prediction set.

$$p^y := \frac{|\{i = 1, \dots, n : y \in S_i\}|}{n} ;$$

remember that the prediction set includes the labels y with $p^y > \epsilon$.

The case where $b_n \le b_i$ for $i < n$ is clearly anomalous: it means that the predicted label $\hat{y}_i$ is at least as sensitive as $\hat{y}_n$ to changes in the postulated label y of x_n. (And Lemmas 2.16 and 2.17 below will show that this case does not happen from some n on almost surely under natural assumptions.) Our implementation of the conformal predictor will simply set $S_i := \mathbb{R}$ in this case. This convention makes our conformal predictor slightly conservative, but it is natural: we do not want to use rays S_i pointing in the wrong direction for making our prediction set narrower.

Let us now find the upper CRR prediction set under the assumption that $b_n > b_i$ for all $i < n$. In this case each set (2.42) is

$$S_i = (-\infty, t_i], \qquad \text{where } t_i := \frac{a_i - a_n}{b_n - b_i} ,$$

except for $S_n := \mathbb{R}$; notice that only $t_1, \dots, t_{n-1}$ are defined. The p-value p^y for any potential label y of x_n is

$$p^y = \frac{|\{i = 1, \dots, n : y \in S_i\}|}{n} = \frac{|\{i = 1, \dots, n-1 : t_i \ge y\}| + 1}{n} .$$

Therefore, the upper CRR prediction set at significance level ϵ is the ray

$$(-\infty, t_{(k_n)}] , \tag{2.43}$$

where $k_n := \lceil (1-\epsilon)n \rceil$ and $t_{(k)}$ stands, as usual, for the kth order statistic (defined in Algorithm 2.2) of $t_1, \dots, t_{n-1}$.

The procedure is summarised as Algorithm 2.2. The algorithm is given a significance level ϵ and outputs the corresponding prediction ray.

The prediction set for the lower CRR is very similar; the main change is that (2.43) becomes

Algorithm 2.3 Lower CRR (simplified)

1: **for** $i = 1, \dots, n-1$
2: **if** $b_n - b_i > 0$ set $t_i := (a_i - a_n)/(b_n - b_i)$
3: **else** set $t_i := -\infty$
4: sort $t_1, \dots, t_{n-1}$ in the ascending order obtaining $t_{(1)} \le \dots \le t_{(n-1)}$
5: output $\left[t_{(\lfloor \epsilon n \rfloor)}, \infty\right)$ as prediction set.

Algorithm 2.4 CRR (simplified)

1: **for** $i = 1, \dots, n-1$
2: **if** $b_n - b_i > 0$ set $u_i := l_i := (a_i - a_n)/(b_n - b_i)$
3: **else** set $l_i := -\infty$ and $u_i := \infty$
4: sort $u_1, \dots, u_{n-1}$ in the ascending order obtaining $u_{(1)} \le \dots \le u_{(n-1)}$
5: sort $l_1, \dots, l_{n-1}$ in the ascending order obtaining $l_{(1)} \le \dots \le l_{(n-1)}$
6: output $\left[l_{(\lfloor (\epsilon/2)n \rfloor)}, u_{(\lceil (1-\epsilon/2)n \rceil)}\right]$ as prediction set.

$$[t_{(\lfloor \epsilon n \rfloor)}, \infty) \ . \tag{2.44}$$

See Algorithm 2.3, in which we drop the first few lines of Algorithm 2.2 introducing the notation.

If we need a bounded prediction interval, we can combine upper and lower CRR and use the conformity scores

$$\alpha_i := \left|\left\{j = 1, \dots, n : e_j \ge e_i\right\}\right| \wedge \left|\left\{j = 1, \dots, n : e_j \le e_i\right\}\right| \ ,$$

where $e_i = y_i - \hat{y}_i$ are the residuals, as before. Notice that the corresponding prediction sets Γ^ϵ at significance level ϵ will satisfy

$$\Gamma^\epsilon \subseteq \Gamma^{\epsilon/2,\text{upper}} \cap \Gamma^{\epsilon/2,\text{lower}} \ , \tag{2.45}$$

where the superscripts "upper" and "lower" indicate the prediction sets output by the upper and lower CRR, respectively. If $\epsilon < 1/2$ and all $n-1$ residuals for the training set are different, the "$\subseteq$" in (2.45) becomes "$=$".

The prediction set for the full CRR defined by the right-hand side of (2.45) is the combination of (2.43) and (2.44):

$$\Gamma^\epsilon = \left[t_{(\lfloor (\epsilon/2)n \rfloor)}, t_{(\lceil (1-\epsilon/2)n \rceil)}\right] \ .$$

The procedure is shown as Algorithm 2.4. Of course, sorting needs to be done just once: since the sequences $u_1, \dots, u_{n-1}$ and $l_1, \dots, l_{n-1}$ are so similar, the sorting operation in line 5 becomes trivial once the sorting operation in line 4 has been carried out.

Let us suppose that the number p of attributes is constant. The computation time of Algorithms 2.2–2.4 is $O(n \log n)$. Indeed,

$$A = (y_1, \dots, y_{n-1}, 0)' - X_n \left[(X_n' X_n + aI_n)^{-1} X_n' (y_1, \dots, y_{n-1}, 0)' \right]$$

and

$$B = (0, \dots, 0, 1)' - X_n \left[(X_n' X_n + aI_n)^{-1} X_n' (0, \dots, 0, 1)' \right]$$

can be computed in time $O(n)$, and the sorting can be done in time $O(n \log n)$ (see, e.g., [55, Part II]).

2.3.3 Two Modifications

The algorithms developed in the previous subsection can be easily modified to allow two alternative ways of computing nonconformity scores. To simplify formulas, we assume $a = 0$ in this subsection (i.e., we will consider least squares). *Conformalized least squares* (CLS) algorithms are CRR algorithms with the ridge parameter a set to 0.

First we consider the case where α_i is defined to be the *deleted residual* $e_{(i)} := y_i - \hat{y}_{(i)}$, where $\hat{y}_{(i)}$ is the least squares prediction for y_i computed from x_i based on the training set $x_1, y_1, \dots, x_{i-1}, y_{i-1}, x_{i+1}, y_{i+1}, \dots, x_n, y_n$. It is well known in statistics that to compute the deleted residuals $e_{(i)}$ we do not need to perform n regressions; they can be easily computed from the usual residuals $e_i = y_i - \hat{y}_i$ by the formula

$$e_{(i)} = \frac{e_i}{1 - h_i} , \tag{2.46}$$

where h_i is the ith diagonal element of the hat matrix H. (For a proof, see [252, Appendix C.7].)

Deleted CLS are defined in the same way as CLS except that the nonconformity scores $\alpha_i := e_i$ are replaced by the deleted versions $\alpha_i := e_{(i)}$. Algorithms 2.2–2.4 will implement the deleted versions if A and B are redefined as follows:

$$a_i := \frac{a_i}{1 - h_i}, \quad b_i := \frac{b_i}{1 - h_i}, \quad i = 1, \dots, n .$$

It is clear that each

$$h_i = x_i' (X_n' X_n)^{-1} x_i$$

can be computed from $(X_n' X_n)^{-1}$ in time $O(1)$ (again assuming that the number p of attributes is constant), and so the deleted CLS can also be implemented in time $O(n \log n)$.

Algorithm 2.5 Studentized CLS (simplified once again)

1: Define a and b by (2.48)
2: **for** $i = 1, \dots, n-1$ set $t_i := (a_i - a_n)/(b_n - b_i)$
3: sort $t_1, \dots, t_{n-1}$ in the ascending order obtaining $t_{(1)} \le \dots \le t_{(n-1)}$
4: output $\left[t_{(\lfloor(\epsilon/2)n\rfloor)}, t_{(\lceil(1-\epsilon/2)n\rceil)}\right]$ as prediction set.

Another natural modification of CLS is half-way between the CLS and the deleted CLS: the nonconformity scores are taken to be

$$\alpha_i := \frac{e_i}{\sqrt{1-h_i}} \tag{2.47}$$

in the case of the upper CLS and its appropriate modifications for the CLS and lower CLS. We will explain the motivation behind this choice momentarily, but first describe how to implement the *studentized CLS* determined by these nonconformity scores. The implementation is just Algorithms 2.2–2.4 with A and B redefined as

$$a_i := \frac{a_i}{\sqrt{1-h_i}}, \quad b_i := \frac{b_i}{\sqrt{1-h_i}}, \quad i = 1, \dots, n\,. \tag{2.48}$$

The computation time is again $O(n \log n)$.

An advantage of the studentized version is that the checks in line 5 in Algorithm 2.2 and line 2 in Algorithm 2.3 become superfluous, since $b_n > b_i$ provided all a_i and b_i are defined (i.e., the denominators in (2.48) are non-zero). This follows from Proposition 7.8 in Chap. 7. Correspondingly, Algorithm 2.4 simplifies to Algorithm 2.5.

Now we explain the standard motivation for studentized CLS. (Remember that this motivation has no bearing on the validity of the constructed conformal predictor; in particular, the smoothed version of this confidence predictor will make errors independently with probability ϵ at each significance level ϵ regardless of whether the assumption of normal noise we are about to make is satisfied or not; cf. Sect. 13.4.) Imagine that the labels y_i are generated from the deterministic objects x_i in the following way:

$$y_i = w \cdot x_i + \xi_i\,, \tag{2.49}$$

where ξ_i are independent normal random variables with the mean 0 and same variance σ^2 (random noise). Set $\xi := (\xi_1, \dots, \xi_n)'$. Since the vector of residuals is $e = (I_n - H_n)Y_n$ (see above), we obtain

$$e = (I_n - H_n)(X_n w + \xi) = (I_n - H_n)\xi \tag{2.50}$$

for any fixed w (the true parameters); therefore, the covariance matrix of the residuals is

$$\text{var}(e) = \text{var}((I_n - H_n)\xi) = (I_n - H_n)\text{var}(\xi)(I_n - H_n)' = \sigma^2(I_n - H_n) \ ,$$

since $\text{var}(\xi) = \sigma^2 I_n$ and $I_n - H_n$ is symmetric and idempotent. We can see that the variance of e_i is $(1 - h_i)\sigma^2$, and the scaling of the residuals e_i by dividing by $\sqrt{1 - h_i}$ will equalize their variances (and even their distributions, since by (2.50), e_i are normally distributed).

Notice that according to our motivational model (2.49) the level of noise ξ_i does not depend on the observed object x_i (the variance of ξ_i remains the same, σ^2). Even in this case, it may be useful to scale residuals. If we suspect that noise can be different in different parts of the object space, heavier scaling may become necessary for satisfactory prediction.

2.3.4 Dual Form Ridge Regression

Least squares and ridge regression procedures can only deal with situations where the number of parameters p is relatively small since they involve inverting a $p \times p$ matrix. They are carried over to high-dimensional problems using the so-called "kernel trick", introduced in machine learning by Vapnik (see, e.g., [365, 366]). (The formulas arrived at by using the kernel trick coincide with those obtained by means of Gaussian processes and reproducing kernel Hilbert spaces; see Sect. 2.9.6.)

We first state the ridge regression procedure in the "dual form". The traditional statistical approach to dualization is to use the easy-to-check matrix equality

$$X_n(X_n'X_n + aI_p)^{-1} = (X_nX_n' + aI_n)^{-1}X_n \ , \tag{2.51}$$

which can be equivalently rewritten as

$$(X_n'X_n + aI_p)^{-1}X_n' = X_n'(X_nX_n' + aI_n)^{-1} \ .$$

(To see that (2.51) is true, multiply it by $(X_nX_n' + aI_n)$ on the left and $(X_n'X_n + aI_p)$ on the right.) Using (2.51), we can rewrite the ridge regression prediction for an object x based on examples $(x_1, y_1, \ldots, x_n, y_n)$ as

$$\hat{y} = Y_n'X_n(X_n'X_n + aI_p)^{-1}x = Y_n'(X_nX_n' + aI_n)^{-1}X_nx \ , \tag{2.52}$$

where the $n \times p$ matrix X_n and vector Y_n are defined as before. The crucial property of the representation $\hat{y} = Y_n'(X_nX_n' + aI_n)^{-1}X_nx$ is that it depends on the objects $x_1, \ldots, x_n, x$ only via the scalar products between them. In particular, if the object space $\mathbf{X}$ is mapped into another Euclidean space (called the *feature space*) $\mathbf{H}$, $F : \mathbf{X} \to \mathbf{H}$ (assumed measurable), and ridge regression is performed in the feature space, the prediction (2.52) can be written in the form

$$\hat{y} = Y_n'(K_n + aI_n)^{-1}k_n \ , \tag{2.53}$$

where K_n is the matrix with elements $(K_n)_{i,j} := \mathcal{K}(x_i, x_j)$, k_n is the vector with elements $(k_n)_i := \mathcal{K}(x, x_i)$, and $\mathcal{K}$ is the *kernel*, defined by

$$\mathcal{K}(x^{(1)}, x^{(2)}) := F(x^{(1)}) \cdot F(x^{(2)}) \tag{2.54}$$

for all $x^{(1)}, x^{(2)} \in \mathbf{X}$. The prediction rule (2.53) is also known as *kernel ridge regression*. The hat matrix in the dual representation is

$$H_n = X_n(X_n'X_n + aI_p)^{-1}X_n' = (X_nX_n' + aI_n)^{-1}X_nX_n' \ ,$$

and if ridge regression is carried out in the feature space, this becomes

$$H_n = (K_n + aI_n)^{-1}K_n \ . \tag{2.55}$$

Now it is easy to represent the CRR algorithms in the kernel form: the only difference from Algorithms 2.2–2.4 is that now C is defined as $I_n - H_n$ with H_n given by (2.55).

The computation time of the kernel form of the CRR algorithms is $O(n^2)$ at the nth step of the online protocol. This can be seen from the well-known (see, e.g., [155, (8)]) and easy-to-check formula

$$\begin{pmatrix} K & k \\ k' & \kappa \end{pmatrix}^{-1} = \begin{pmatrix} K^{-1} + dK^{-1}kk'K^{-1} & -dK^{-1}k \\ -dk'K^{-1} & d \end{pmatrix} \ , \tag{2.56}$$

where K is a square matrix, k a vector, κ a number, and

$$d := \frac{1}{\kappa - k'K^{-1}k} \tag{2.57}$$

(formula (2.56) assumes that the denominator in (2.57) is different from 0 so that d is defined and, of course, that K^{-1} is defined). Indeed, by this formula $(K_n + aI_n)^{-1}$ can be updated from the previous trial of the online learning protocol in time $O(n^2)$, and both

$$A = (y_1, \dots, y_{n-1}, 0)' - (K_n + aI_n)^{-1}\left[K_n(y_1, \dots, y_{n-1}, 0)'\right]$$

and

$$B = (0, \dots, 0, 1)' - (K_n + aI_n)^{-1}\left[K_n(0, \dots, 0, 1)'\right]$$

can be computed in time $O(n^2)$. We mentioned some conditions on the validity of formula (2.56), but those conditions are satisfied in our context, at least when $a > 0$:

the theorem on normal correlation (see, e.g., [329, Theorem 2.13.2]) implies that d is well-defined (and positive) whenever the matrix $\left(\begin{smallmatrix} K & k \\ k' & \kappa \end{smallmatrix}\right)$ is positive definite; the latter condition is satisfied when the formula is used for updating $(K_n + aI_n)^{-1}$, $a > 0$, to $(K_{n+1} + aI_{n+1})^{-1}$ (in which case $K = K_n + aI_n$, $k = k_n$, and $\kappa = \mathcal{K}(x_n, x_n) + a$).

The above construction also works in the case where $\mathbf{H}$ is an arbitrary Hilbert space. This is especially obvious in the case of a finite horizon, where only the linear span of the observed objects matters.

The essence of the kernel trick is that one does not need to consider the feature space in explicit form [366, Sect. 10.5.2]. It is clear that any kernel (2.54) is symmetric,

$$\mathcal{K}(x^{(1)}, x^{(2)}) = \mathcal{K}(x^{(2)}, x^{(1)}), \quad \forall x^{(1)}, x^{(2)} \in \mathbf{X}\,,$$

and nonnegative definite,

$$\sum_{i=1}^{m} \sum_{j=1}^{m} \mathcal{K}(x^{(i)}, x^{(j)}) a_i a_j \geq 0 \tag{2.58}$$

for all $x^{(1)}, \ldots, x^{(m)} \in \mathbf{X}$ and all $a_1, \ldots, a_m \in \mathbb{R}$. (To see that (2.58) is true, notice that

$$\begin{aligned} &\left\| a_1 F(x^{(1)}) + \cdots + a_m F(x^{(m)}) \right\|^2 \\ &= \left(a_1 F(x^{(1)}) + \cdots + a_m F(x^{(m)}) \right) \cdot \left(a_1 F(x^{(1)}) + \cdots + a_m F(x^{(m)}) \right) \geq 0\,.) \end{aligned}$$

It turns out that the opposite statement is also true if we ignore the requirement that the feature mapping be measurable: any function $\mathcal{K} : \mathbf{X}^2 \to \mathbb{R}$ that is symmetric and nonnegative definite can be represented in the form (2.54) (see, e.g., [341, Theorem 4.16]). In interesting cases, the feature mapping F to a Hilbert space $\mathbf{H}$ can be chosen measurable.

2.4 Conformalized Nearest Neighbours Regression

Least squares and ridge regression are just two of the standard regression algorithms; conformal predictors can be implemented in a feasible way for nonconformity measures based on many other regression algorithms. Such an implementation is especially simple in the case of the nearest neighbours algorithm.

The idea of the k-nearest neighbours algorithms, where $k \in \mathbb{N}$ is the number of "neighbours" taken into account, is as follows. Suppose the object space $\mathbf{X}$ is a metric space (for example, the usual Euclidean distance is often used if $\mathbf{X} = \mathbb{R}^p$). To give a prediction for a new object x, find the k objects $x_{i_1}, \ldots, x_{i_k}$ among the

known examples that are nearest to x in the sense of the chosen metric (assuming, for simplicity, that there are no ties). In the problem of classification, the predicted label $\hat{y}$ of x is obtained by "voting": it is defined to be the most frequent label among $y_{i_1}, \dots, y_{i_k}$. In regression, we can take, e.g., the mean or the median of $y_{i_1}, \dots, y_{i_k}$.

We will only consider the version of the k-nearest neighbours regression (k-NNR) where the prediction $\hat{y}$ for a new object x based on the training set (x_i, y_i), $i = 1, \dots, n$, is defined to be the arithmetic mean of the labels of the k nearest neighbours of x among $x_1, \dots, x_n$. It will be easy to see that the more robust procedure where arithmetic mean is replaced by median also leads to a feasible conformal predictor.

Let us consider, in the context of (2.33), the nonconformity scores $\alpha_i := y_i - \hat{y}_{(i)}$, where $\hat{y}_{(i)}$ is the k-NNR prediction for x_i based on the training set (x_j, y_j), $j = 1, \dots, i-1, i+1, \dots, n$. The conformal predictor determined by this nonconformity measure (*k-NNR conformal predictor*) is implemented by the upper CRR algorithm (Algorithm 2.2) with the only modification that a_i and b_i are now defined as follows (we assume that $n > k$ and that all distances between the objects are different):

- a_n is the minus arithmetic mean of the labels of x_n's k nearest neighbours and $b_n = 1$;
- if $i < n$ and x_n is among the k nearest neighbours of x_i, a_i is x_i's label minus the arithmetic mean of the labels of those nearest neighbours with x_n's label set to 0, and $b_i = -1/k$;
- if $i < n$ and x_n is not among the k nearest neighbours of x_i, a_i is x_i's label minus the arithmetic mean of the labels of x_i's k nearest neighbours, and $b_i = 0$.

The nonconformity score for the ith example is $\alpha_i = a_i + b_i y$, where y is the label for x_n that is being tried.

Plugging the values a_i and b_i as defined in the previous paragraph into Algorithms 2.2, 2.3, and 2.4, we obtain *upper conformalized nearest neighbours*, *lower conformalized nearest neighbours*, and *conformalized nearest neighbours*, respectively. In none of the NNR-based algorithms there is need to check $b_n - b_i > 0$ (see, e.g., line 5 in Algorithm 2.2), since now we always have $b_n - b_i = 1 + 1/k$.

Instead of measuring distance in the original example space $\mathbf{X}$ we can measure it in the feature space, which corresponds to using the function

$$\begin{aligned}\rho(x^{(1)}, x^{(2)}) &:= \left\| F(x^{(1)}) - F(x^{(2)}) \right\|^2 \\ &= \left(F(x^{(1)}) - F(x^{(2)}) \right) \cdot \left(F(x^{(1)}) - F(x^{(2)}) \right) \\ &= \mathcal{K}(x^{(1)}, x^{(1)}) + \mathcal{K}(x^{(2)}, x^{(2)}) - 2\mathcal{K}(x^{(1)}, x^{(2)})\end{aligned}$$

as the distance (remember that in the case of k-NNR conformal prediction we are only interested in the distance up to a monotonic transformation).

2.5 Efficiency of Conformalized Ridge Regression

In this section we describe and start implementing in a special case what we will call the *Burnaev–Wasserman programme* (we learned the idea of this research programme from Evgeny Burnaev and Larry Wasserman). The specific conformal predictors constructed so far were obtained by "conformalizing" various underlying algorithms, which are typically chosen from among standard prediction algorithms used in machine learning. This will remain our main strategy in the rest of the book. The Burnaev–Wasserman programme concerns the efficiency of this strategy.

Let us start from conformalizing Bayesian prediction algorithms. Suppose we know, or postulate, the true probability distribution generating the data, and it is a power distribution. With a known data-generating distribution, we can often find the optimal prediction algorithm, the *Bayes algorithm*, under this distribution and for natural definitions of optimality. In this chapter we are interested in Bayes algorithms outputting prediction sets, which we will call *Bayesian predictions* (an example will be given in Sect. 2.5.2). The Bayesian predictions satisfy strong properties of validity, but they are guaranteed only under the known or postulated probability distribution.

A *set predictor* is a measurable function

$$\Gamma : \mathbf{Z}^* \times \mathbf{X} \to 2^{\mathbf{Y}} . \tag{2.59}$$

It is required to be measurable in the same sense as the confidence predictor (2.7) in Sect. 2.1.2; therefore, a set predictor is a confidence predictor that does not depend on ϵ.

To obtain a set predictor that is valid under the exchangeability model, we can run the conformal predictor on top of the Bayes algorithm, "conformalizing" the latter. Namely, we design a nonconformity measure based on the Bayes algorithm (or directly on the data-generating distribution) and then feed it into the recipe for conformal prediction (or one of its modifications discussed in Chap. 4).

The conformalizing procedure improves the guaranteed property of validity in an important respect: now we have validity under the exchangeability model. This may be important when the data-generating distribution is postulated rather than known, which is usually the case in practice. The main question asked in the Burnaev–Wasserman programme is whether we have to pay in terms of efficiency, under the postulated data-generating distribution, for the gain in validity. If the price is low, conformalizing is akin to buying a cheap insurance policy to protect us against the postulated data-generating distribution being wrong. We will discuss the insurance metaphor in a different but similar context in Sect. 10.1. If the price of protection turns out to be high, this might mean that the Bayesian assumption is too fragile.

Our discussion so far is also applicable to narrow parametric models in place of Bayesian models, first of all Gaussian models. There may be nearly optimal prediction algorithms under such models, and then the task is to explore whether

we lose much in efficiency, under those models, when conformalizing those nearly optimal algorithms.

2.5.1 Hard and Soft Models

In our informal discussions in the framework of the Burnaev–Wasserman programme, we will sometimes distinguish between hard and soft models. Despite some common features, this distinction is very different from that between null and alternative hypotheses in the Neyman–Pearson theory of hypothesis testing. In particular, we are often interested in the case where the hard model (almost invariably the exchangeability or randomness model) is more general than the soft model (which may be a Bayesian model).

The *hard models* are the usual statistical models: our working hypothesis is that the dataset was generated by one of the probability distributions in the model. In particular, the validity of our confidence predictors is allowed to depend on the hard model.

In addition to the accepted hard model, one often has other a priori information about the data-generating distribution: e.g., only a few parameters might provide the bulk of the information relevant to prediction. Whereas we might hesitate to include such a priori information in the hard model explicitly, since it might destroy the validity of our confidence predictor if this information happened to be far from the truth, we might still be able to use such information, comprising our *soft model*, in designing accurate confidence predictors.

In this section our hard model will be the exchangeability model, and our soft model will be Bayesian. Bayesian confidence predictors are only valid under the soft model, and therefore, we will be interested in the corresponding conformal predictors. Whether the soft model is true or not affects only the efficiency, but not validity, of the conformal predictors.

Separation of the available information about the data-generating distribution into the hard model and soft model increases robustness of confidence predictors with respect to modelling errors. If such an error occurs in the soft model, the validity of predictions is not affected. At worst the predictions will become useless, but they will not become misleading (with high probability under any distribution in the hard model).

2.5.2 Bayesian Ridge Regression

In this subsection we give only a brief summary of Bayesian ridge regression; see Sect. 13.4.1 for details. We are interested in the case where the number $n - 1$ of training examples is large, whereas the number p of attributes is fixed.

We assume that, given the objects, the labels $y_1, \dots, y_n$ are generated by the rule (2.49), where w is a random vector now distributed as $\mathbf{N}(0, (\sigma^2/a)I)$ ($I := I_p$ being the unit $p \times p$ matrix), each ξ_i is distributed as $\mathbf{N}(0, \sigma^2)$, the random elements $w, \xi_1, \dots, \xi_n$ are independent (given the objects), and σ and a are given positive numbers.

The conditional distribution for the label y_n of the test object x_n given the training set (x_i, y_i), $i = 1, \dots, n-1$, and x_n is

$$\mathbf{N}\left(\hat{y}_n, (1 + g_n)\sigma^2\right)$$

(see (13.27)), where

$$\begin{aligned} \hat{y}_n &:= x_n'(X'X + aI)^{-1}X'Y \,, \\ g_n &:= x_n'(X'X + aI)^{-1}x_n \,, \end{aligned} \tag{2.60}$$

$X = X_{n-1}$ is the data matrix for the training set (the $(n-1) \times p$ matrix whose ith row is x_i', $i = 1, \dots, n-1$), and $Y = Y_{n-1}$ is the vector $(y_1, \dots, y_{n-1})'$ of the training labels. Therefore, the *Bayesian prediction interval* is

$$(B_*, B^*) := \left(\hat{y}_n - \sqrt{1 + g_n}\sigma \mathbf{z}_{\epsilon/2}, \hat{y}_n + \sqrt{1 + g_n}\sigma \mathbf{z}_{\epsilon/2}\right) , \tag{2.61}$$

where ϵ is the significance level (the permitted probability of error, so that $1 - \epsilon$ is the required coverage probability) and $\mathbf{z}_{\epsilon/2}$ is the $(1 - \epsilon/2)$-quantile of the standard normal distribution $\mathbf{N}_{0,1}$.

The prediction interval (2.61) enjoys several desiderata: it is unconditionally valid, in the sense that its error probability is equal to the given significance level ϵ; it is also valid conditionally on the training set and the test object x_n; finally, this prediction interval is the shortest possible conditionally valid interval. We will refer to the class of algorithms producing prediction intervals (2.61) (and depending on the parameters σ and a) as *Bayesian ridge regression* (BRR).

2.5.3 *Efficiency of CRR*

In the rest of this section, the hard model is, as already mentioned, the exchangeability model, and the soft model assumes that the objects x_i are IID while y_i are generated from (2.49), $w \sim \mathbf{N}(0, (\sigma^2/a)I)$, and $\xi_i \sim \mathbf{N}(0, \sigma^2)$, all independent. (It is slightly more natural to consider the exchangeability model rather than the randomness model as our hard model since a typical element of the soft model is an exchangeable but not power distribution.) In this subsection we will see that under this soft model complemented by natural (and standard) assumptions CRR is asymptotically close to BRR (2.61) (optimized under the soft model), and therefore

is approximately conditionally valid and efficient. On the other hand, we have the unconditional validity of CRR under the exchangeability assumption, regardless of whether (2.49) holds.

We assume an infinite sequence of examples $(x_1, y_1), (x_2, y_2), \ldots$ but consider only the first n of them and let $n \to \infty$. We make both the randomness assumption about the objects $x_1, x_2, \ldots$ (the objects are generated independently from the same distribution) and the assumption (2.49); however, we can relax the assumption that w is distributed as $\mathbf{N}(0, (\sigma^2/a)I)$. These are all the assumptions used in our main result:

(A1) The random objects $x_i \in \mathbb{R}^p$, $i = 1, 2, \ldots$, are IID.
(A2) The second-moment matrix $\mathbb{E}(x_1 x_1')$ of x_1 exists and is non-singular.
(A3) The random vector $w \in \mathbb{R}^p$ is independent of $x_1, x_2, \ldots$.
(A4) The labels $y_1, y_2, \ldots$ are generated by $y_i = w \cdot x_i + \xi_i$, where ξ_i are Gaussian noise variables distributed as $\mathbf{N}(0, \sigma^2)$ and independent between themselves, of the objects x_i, and of w.

Notice that the assumptions imply that the random examples (x_i, y_i), $i = 1, 2, \ldots$, are IID given w. It will be clear from the proof that the assumptions can be relaxed further (but we have tried to make them as simple as possible).

Theorem 2.7 *Under the assumptions (A1)–(A4), the prediction sets output by CRR are intervals from some n on almost surely, and the differences between the upper and lower end-points of the prediction intervals for BRR and CRR are asymptotically Gaussian:*

$$\frac{\sqrt{n}}{\sigma}(B^* - C^*) \xrightarrow{\text{law}} \mathbf{N}\left(0, \frac{(\epsilon/2)(1-\epsilon/2)}{\phi^2(\mathbf{z}_{\epsilon/2})} - \mu' \Sigma^{-1} \mu\right), \tag{2.62}$$

$$\frac{\sqrt{n}}{\sigma}(B_* - C_*) \xrightarrow{\text{law}} \mathbf{N}\left(0, \frac{(\epsilon/2)(1-\epsilon/2)}{\phi^2(\mathbf{z}_{\epsilon/2})} - \mu' \Sigma^{-1} \mu\right), \tag{2.63}$$

where ϕ is the density of the standard normal distribution $\mathbf{N}_{0,1}$, $\mu := \mathbb{E}(x_1)$ is the expectation of x_1, and $\Sigma := \mathbb{E}(x_1 x_1')$ is the second-moment matrix of x_1.

The theorem will be proved in Sect. 2.8.2, and in the rest of this subsection we will discuss it. We can see from (2.62) and (2.63) that the symmetric difference between the prediction intervals output by BRR and CRR shrinks to 0 as $O(n^{-1/2})$ in Lebesgue measure with high probability.

Let us first see what the typical values of the standard deviation (the square root of the variance) in (2.62) and (2.63) are. The second term in the variance does not affect it significantly since $0 \le \mu' \Sigma^{-1} \mu \le 1$. Indeed, denoting the covariance matrix of x_1 by C and using the Sherman–Morrison formula (see Remark 2.8 below), we have:

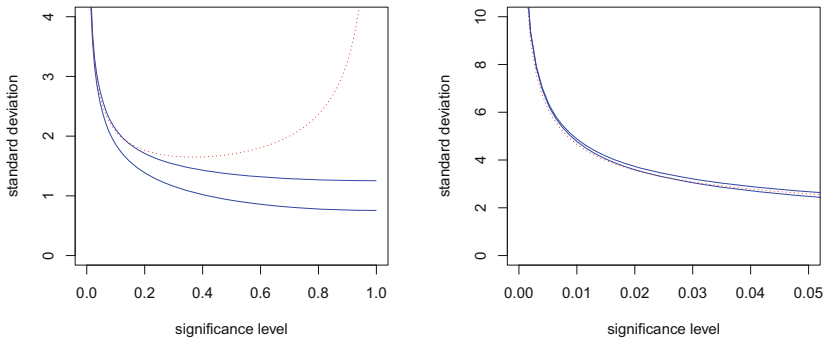


Fig. 2.1 The limits for the standard deviation in Theorem 2.7 as a function of the significance level $\epsilon \in (0, 1)$ (left) and $\epsilon \in (0, 0.05]$ (right) shown as solid blue lines; the asymptotic expression in (2.65) is shown as a dotted red line. In all cases $\sigma = 1$

$$\mu' \Sigma^{-1} \mu = \mu' (C + \mu\mu')^{-1} \mu = \mu' \left(C^{-1} - \frac{C^{-1} \mu\mu' C^{-1}}{1 + \mu' C^{-1} \mu} \right) \mu$$
$$= \mu' C^{-1} \mu - \frac{(\mu' C^{-1} \mu)^2}{1 + \mu' C^{-1} \mu} = \frac{\mu' C^{-1} \mu}{1 + \mu' C^{-1} \mu} \in [0, 1] \tag{2.64}$$

(we write $[0, 1]$ rather than $(0, 1)$ because C is permitted to be singular.[2]) The first term, on the other hand, can affect the variance more significantly, and the significant dependence of the variance on ϵ is natural: the accuracy of the conformal predictor degrades for small ϵ since, instead of using all data for estimating the endpoints of the prediction interval, it relies, under the exchangeability model, on the scarcer information provided by examples in the tails of the distribution generating the labels. Figure 2.1 illustrates the dependence of the standard deviation of the asymptotic distribution on ϵ. The upper solid line in it corresponds to $\mu' \Sigma^{-1} \mu = 0$, and the lower solid line corresponds to $\mu' \Sigma^{-1} \mu = 1$. The possible values for the standard deviation lie between these two lines. The asymptotic behaviour of the standard deviation as $\epsilon \to 0$ is given by

$$\sqrt{\epsilon(1 - \epsilon/2)\pi \exp(\mathbf{z}_{\epsilon/2}^2) - \theta} \sim (-\epsilon \ln \epsilon)^{-1/2} \tag{2.65}$$

uniformly in $\theta \in [0, 1]$.

The assumptions (A1)–(A4) do not involve a, and Theorem 2.7 continues to hold if we set $a := 0$. Theorem 2.7 can thus also be considered as an efficiency result about conformalizing the standard non-Bayesian least squares procedure; this procedure outputs precisely (B_*, B^*) with $a := 0$ as its prediction intervals

[2] If C is singular, we can apply (2.64) to $\Sigma_\epsilon := \Sigma + \epsilon I$ and $C_\epsilon := C + \epsilon I$, where $\epsilon > 0$, in place of Σ and C, respectively, and let $\epsilon \to 0$.

(see, e.g., [313, Sect. 5.3.1]). The least squares procedure has guaranteed coverage probability under weaker assumptions than BRR (not requiring assumptions about w); however, its validity is not conditional, similarly to CRR.

Remark 2.8 The Sherman–Morrison formula (see, e.g., [155, (3)]) is

$$(K + uv')^{-1} = K^{-1} - \frac{K^{-1}uv'K^{-1}}{1 + v'K^{-1}u} , \tag{2.66}$$

where K is a square matrix and u and v are vectors. It can be checked easily by multiplying the right-hand side by $K + uv'$ on the left and simplifying.

2.6 Are There Other Ways to Achieve Validity?

In this section we will see that conformal predictors are essentially the only confidence predictors in very natural classes that satisfy our non-asymptotic properties of validity.

Let us say that a confidence predictor is *invariant* if $\Gamma^\epsilon(z_1, \dots, z_{n-1}, x_n)$ does not depend on the order in which $z_1, \dots, z_{n-1}$ are listed. Since we assume exchangeability, the invariant confidence predictors constitute a natural class (see, e.g., the description of the "sufficiency principle" in [64]; later in this book, however, we will also study confidence predictors that are not invariant, such as inductive conformal predictors in Chap. 4).

If Γ_1 and Γ_2 are (deterministic) confidence predictors, we will say that Γ_1 is *at least as good as* Γ_2 if, for any $\epsilon \in (0, 1)$, any n, and and $x_1, y_1, x_2, y_2, \dots$,

$$\Gamma_1^\epsilon (x_1, y_1, \dots, x_{n-1}, y_{n-1}, x_n) \subseteq \Gamma_2^\epsilon (x_1, y_1, \dots, x_{n-1}, y_{n-1}, x_n) .$$

We will also express this relation by saying that Γ_1 *dominates* Γ_2.

We have given these definitions in the context of online predictors, but they are also applicable to one-off confidence predictors: just remove the quantifier over n. As a gentle introduction, we start from a result about one-off predictors.

Proposition 2.9 *Any one-off confidence predictor that is conservatively valid under exchangeability is dominated by a weak conformal predictor. If, additionally, it is invariant, then it is dominated by a conformal predictor.*

Proof Let n be the size of a conservatively valid weak conformal predictor Γ. Define $S(z_1, \dots, z_n)$ as the infimum of ϵ such that Γ makes an error when fed with $z_1, \dots, z_{n-1}$ as the training set and z_n's object as the test object. Let us check that the weak conformal predictor $\tilde{\Gamma}$ determined by S as conformity statistic is at least as good as Γ.

For any sequence $(z_1, \dots, z_n) \in \mathbf{Z}^n$, let P be the uniform probability measure on all permutations of $(z_1, \dots, z_n)$, and let $\epsilon \in (0, 1)$. Since P is exchangeable, Γ's

property of validity implies that the fraction of the permutations at which $S \le \epsilon$ is at most ϵ (this is true with any $\epsilon' > \epsilon$ in place of ϵ, and so it is true for ϵ as well). Therefore, $\tilde{\Gamma}$ is at least as good as Γ.

The following result asserts that, in the online mode of prediction, invariant conservatively valid confidence predictors are conformal predictors or can be improved to become conformal predictors.

Theorem 2.10 *Suppose* $\mathbf{Z}$ *is a standard Borel space. Let* Γ *be an invariant conservatively valid confidence predictor. Then there is a conformal predictor that is at least as good as* Γ.

Theorem 2.10 will be proved in Sect. 2.8.3, and the proof will be based on the same idea as the proof of Proposition 2.9. It is interesting that the proof will not use the fact that the random variables $\xi_1^{(\epsilon)}, \xi_2^{(\epsilon)}, \dots$ from the definition of Γ's conservative validity are independent. The independence then appears automatically as property of the errors of (or, as explained in the next section, of the p-values produced by) the resulting conformal predictor.

2.7 Conformal Transducers

There are two convenient ways to represent a conformal predictor: as a confidence predictor and as a "confidence transducer". So far we have been using the first way; the goal of this section is to introduce the second way, which is simpler mathematically, and to discuss connections between the two. This will be particularly needed in Chap. 7 and Part III.

2.7.1 Definitions and Properties of Validity

A *randomized confidence transducer* is a function f mapping the nonempty sequences in $(\mathbf{X} \times [0,1] \times \mathbf{Y})^*$ to $[0,1]$. It is called "transducer" because it can be regarded as mapping each input sequence $(x_1, \tau_1, y_1, x_2, \tau_2, y_2, \dots)$ in $(\mathbf{X} \times [0,1] \times \mathbf{Y})^\infty$ into the output sequence $(p_1, p_2, \dots)$ of *p-values* defined by $p_n := f(x_1, \tau_1, y_1, \dots, x_n, \tau_n, y_n)$, $n = 1, 2, \dots$; we often drop "confidence" in "confidence transducer". We say that f is an *exactly valid randomized transducer* (or just *exact randomized transducer*) if the output p-values $p_1 p_2 \dots$ are always distributed according to the uniform distribution $\mathbf{U}^\infty$ on $[0,1]^\infty$, provided the input examples $z_n = (x_n, y_n)$, $n = 1, 2, \dots$, are generated by an exchangeable probability distribution on $\mathbf{Z}^\infty$ and the numbers $\tau_1, \tau_2, \dots$ are generated independently from the uniform distribution $\mathbf{U}$ on $[0,1]$.

We can extract exact randomized transducers from nonconformity measures: given a nonconformity measure A, for each nonempty sequence

$$(x_1, \tau_1, y_1, \dots, x_n, \tau_n, y_n) \in (\mathbf{X} \times [0,1] \times \mathbf{Y})^*$$

define

$$f(x_1, \tau_1, y_1, \dots, x_n, \tau_n, y_n) := \frac{|\{i : \alpha_i > \alpha_n\}| + \tau_n |\{i : \alpha_i = \alpha_n\}|}{n} , \tag{2.67}$$

where $i = 1, \dots, n$ and α_i, $i = 1, 2, \dots$, are computed from $z_i = (x_i, y_i)$ using A by the usual (cf. (2.23)) formula

$$\alpha_i := A(\lbag z_1, \dots, z_n \rbag, z_i) . \tag{2.68}$$

Each randomized transducer f that can be obtained in this way will be called a *smoothed conformal transducer*.

Proposition 2.11 *Each smoothed conformal transducer is an exact randomized transducer.*

This proposition is a special case of Theorem 11.1, which will be proved in Chap. 11 (see Sects. 11.2.3 and 11.5.1).

In a similar way we can define (deterministic) *conformal transducers* f: given a nonconformity measure A, for each nonempty sequence $(z_1, \dots, z_n) \in \mathbf{Z}^*$ set

$$f(z_1, \dots, z_n) := \frac{|\{i : \alpha_i \geq \alpha_n\}|}{n} ,$$

where α_i are computed by (2.68) as before. In general, a (deterministic) *transducer* is a function f of the type $\mathbf{Z}^* \setminus \{\square\} \to [0,1]$, $\square$ being the empty sequence; as before, we associate with f a mapping from $z_1 z_2 \dots$ to the *p-values* $p_1 p_2 \dots$ ($p_n := f(z_1, \dots, z_n)$). We say that f is a *conservatively valid transducer* (or *conservative transducer*) if there exists a sequence ξ_n, $n = 1, 2, \dots$, of $[0,1]$-valued random variables on the extended probability space (2.11) such that:

- for each n, ξ_n only depends on the first n examples;
- for each n, $p_n \leq \xi_n$;
- the sequence $\xi_1, \xi_2, \dots$ is distributed as $\mathbf{U}^\infty$, provided the examples are generated from an exchangeable distribution P on $\mathbf{Z}^\infty$.

The following implication of Proposition 2.11 is obvious:

Corollary 2.12 *Each conformal transducer is conservative.*

We can fruitfully discuss confidence transducers even in the case of general example spaces $\mathbf{Z}$, not necessarily products $\mathbf{X} \times \mathbf{Y}$ of object and label spaces. But in the latter case we can associate a confidence predictor $\Gamma = f'$ with each confidence transducer f defining (2.21) as

$$\{y \in \mathbf{Y} : f(x_1, y_1, \dots, x_{n-1}, y_{n-1}, x_n, y) > \epsilon\} . \tag{2.69}$$

Vice versa, with any confidence predictor Γ we can associate the confidence transducer $f = \Gamma'$ defined by

$$f(x_1, y_1, \dots, x_n, y_n) := \sup \left\{ \epsilon : y_n \in \Gamma^\epsilon(x_1, y_1, \dots, x_{n-1}, y_{n-1}, x_n) \right\} . \tag{2.70}$$

Letting x_i in (2.69) and (2.70) range over the extended object space $\mathbf{X} \times [0, 1]$, we obtain the definition of the randomized confidence predictor f' associated with a randomized confidence transducer f and the definition of the randomized confidence transducer Γ' associated with a randomized confidence predictor Γ.

We will see in the next subsection that the expositions of the theory of hedged prediction in terms of conformal transducers and conformal predictors are essentially equivalent. But first we slightly strengthen Proposition 2.4.

A randomized confidence predictor is *strongly exact* if, for any exchangeable probability distribution on $\mathbf{Z}^\infty$ and any sequence $(\epsilon_1, \epsilon_2, \dots) \in (0, 1)^\infty$ of significance levels, the sequence of random variables $\operatorname{err}_n^{\epsilon_n}(\Gamma)$, $n = 1, 2, \dots$, is distributed as the product $\mathbf{B}_{\epsilon_1} \times \mathbf{B}_{\epsilon_2} \times \dots$ of Bernoulli distributions with parameters $\epsilon_1, \epsilon_2, \dots$. It can be defined in a similar way what it means for a confidence predictor Γ to be strongly conservative.

Theorem 11.1 will also imply the following proposition.

Proposition 2.13 *Any smoothed conformal predictor is strongly exact.*

2.7.2 Normalized Confidence Predictors and Confidence Transducers

To obtain full equivalence between confidence transducers and confidence predictors, a further natural restriction has to be imposed on the latter: they will be required to be "normalized". This is a mild restriction since each confidence predictor can be normalized in such a way that its quality does not suffer.

Formally, the *normal form* Γ_{norm} of a confidence predictor Γ is defined by

$$\Gamma^\epsilon_{\text{norm}}(x_1, y_1, \dots, x_n) := \bigcup_{\epsilon' > \epsilon} \Gamma^{\epsilon'}(x_1, y_1, \dots, x_n) .$$

We say that Γ is *normalized* if $\Gamma_{\text{norm}} = \Gamma$. These definitions are also applicable to randomized confidence predictors, in which case x_i range over the extended object space $\mathbf{X} \times [0, 1]$. The following proposition lists some basic properties of the operation $_{\text{norm}}$ and normalized confidence predictors.

Proposition 2.14 *All conformal predictors and smoothed conformal predictors are normalized. For any confidence predictor Γ (randomized or deterministic):*

1. Γ_{norm} *is* at least as good as Γ, *in the sense that*

$$\Gamma^{\epsilon}_{\text{norm}}(x_1, y_1, \ldots, x_n) \subseteq \Gamma^{\epsilon}(x_1, y_1, \ldots, x_n)$$

   for all $x_1, y_1, \ldots, x_n$ *and* ϵ*;*
2. $(\Gamma_{\text{norm}})_{\text{norm}} = \Gamma_{\text{norm}}$*;*
3. Γ *is normalized if and only if the set*

$$\left\{\epsilon : y_n \in \Gamma^{\epsilon}(x_1, y_1, \ldots, x_n)\right\}$$

   is open in $(0, 1)$ *for all* $x_1, y_1, \ldots, x_n$*;*
4. *If* Γ *is exact (resp. conservative, resp. strongly exact), then* Γ_{norm} *is exact (resp. conservative, resp. strongly exact).*

Proof All (smoothed) conformal predictors are normalized because all inequalities involving ϵ in (2.22) and (2.24) are strict. Properties 1 and 2 are obvious. Property 3 follow from the following restatement of the definition of Γ_{norm}:

$$y_n \in \Gamma^{\epsilon}_{\text{norm}}(x_1, y_1, \ldots, x_n)$$

if and only if

$$\exists \epsilon' > \epsilon : y_n \in \Gamma^{\epsilon'}(x_1, y_1, \ldots, x_n) \ .$$

Property 4 follows from

$$\text{err}^{\epsilon}_n(\Gamma_{\text{norm}}) = \inf_{\epsilon' > \epsilon} \text{err}^{\epsilon'}_n(\Gamma) \ .$$

□

The next proposition asserts the equivalence of confidence transducers and normalized confidence predictors (in particular, the equivalence of conformal transducers and conformal predictors). Remember that f' is the confidence predictor associated with a confidence transducer f, and Γ' is the confidence transducer associated with a confidence predictor Γ.

Proposition 2.15 *For each confidence transducer* f*, the confidence predictor* f' *is normalized and* $f'' = f$*. For each normalized confidence predictor* Γ*,* $\Gamma'' = \Gamma$*. If* $\Gamma = f'$ *is the normalized confidence predictor associated with a confidence transducer* $f = \Gamma'$*,*

$$\text{err}^{\epsilon}_n(\Gamma, (z_1, z_2, \ldots)) = 1_{f(z_1, \ldots, z_n) \le \epsilon} \tag{2.71}$$

for any data sequence $z_1, z_2, \ldots$*. If a randomized confidence transducer* f *is exact,* f' *is strongly exact. If a randomized confidence predictor* Γ *is strongly exact,* Γ'

is exact. If f is a (smoothed) conformal transducer, f' is a (smoothed) conformal predictor. If Γ is a (smoothed) conformal predictor, Γ' is a (smoothed) conformal transducer.

Proof Most of the statements in the proposition are obvious. Equality (2.71) follows from the definition of f'. This equality implies that the confidence transducer Γ' is exact for any strongly exact randomized confidence predictor Γ. Indeed, the p-values p_n output by Γ' have the uniform distribution $\mathbf{U}$ on $[0,1]$, and it remains to apply the following simple fact: if a sequence $p_1, p_2, \dots$ of random variables distributed as $\mathbf{U}$ is such that, for any sequence $(\epsilon_1, \epsilon_2, \dots) \in (0,1)^\infty$, the random variables $1_{p_n \le \epsilon_n}$, $n = 1, 2, \dots$, are independent, then the random variables p_n themselves are independent (see, e.g., [329], the theorem in Sect. 2.5, p. 215).

2.8 Proofs

2.8.1 Proof of Theorem 2.2

We may assume that the examples $z_1, \dots, z_{l+1}$ are generated from a power distribution Q^{l+1} such that the probability distribution Q on $\mathbf{Z}$ is concentrated on the set $\{(x, y^{(1)}), (x, y^{(2)})\} \subseteq \mathbf{Z}$, for some arbitrarily fixed $x \in \mathbf{X}$ and distinct $y^{(1)}, y^{(2)} \in \mathbf{Y}$ (remember that we assumed $|\mathbf{X}| \ge 1$ and $|\mathbf{Y}| > 1$). Therefore, we assume, without loss of generality, that $\mathbf{Z} = \{0,1\}$. Set $n := l+1$ and suppose that $\mathrm{err}_n^\epsilon(\Gamma) = 1$ with probability ϵ for all $\epsilon \in (0,1)$.

For each k let $f(k)$ be the probability that $\mathrm{err}_n^\epsilon(\Gamma, (z_1, \dots, z_n)) = 1$ where $(z_1, \dots, z_n) \in \{0,1\}^n$ is drawn from the uniform distribution on the set of all binary sequences of length n with k 1s. Since the expected value of $f(k)$ is ϵ w.r. to any binomial distribution on $\{0, 1, \dots, n\}$, the standard completeness result (see, e.g., [222, Example 4.3.1]) implies that $f(k) = \epsilon$ for all $k = 0, 1, \dots, n$. Therefore, $\epsilon \binom{n}{k}$ is an integer for all k and ϵ, which cannot be true.

2.8.2 Proof of Theorem 2.7

For concreteness, we concentrate on the convergence (2.62) for the upper end-points of the conformal and Bayesian prediction intervals, which we rewrite as

$$\sqrt{n}(B^* - C^*) \xrightarrow{\text{law}} \mathbf{N}\left(0, \frac{\alpha(1-\alpha)}{f^2(\zeta_\alpha)} - \sigma^2 \mu' \Sigma^{-1} \mu\right), \tag{2.72}$$

where $\alpha := 1 - \epsilon/2$, f is the probability density of $\mathbf{N}(0, \sigma^2)$, and $\zeta_\alpha := \mathbf{z}_{\epsilon/2}\sigma$ is the α-quantile of $\mathbf{N}(0, \sigma^2)$. We split the proof into a series of steps.

Regularizing the Rays in Upper CRR

The upper CRR looks difficult to analyze in general, since the sets (2.42) may be rays pointing in the opposite directions. Fortunately, the awkward case $b_n \le b_i$ $(i < n)$ will be excluded for large

n under our assumptions (see Lemma 2.17 below). The following lemma gives a simple sufficient condition for its absence.

Lemma 2.16 *Suppose that, for each* $c \in \mathbb{R}^p \setminus \{0\}$,

$$(c \cdot x_n)^2 < \sum_{i=1}^{n-1} (c \cdot x_i)^2 + a \|c\|^2 \ , \tag{2.73}$$

where $\|\cdot\|$ *stands for the Euclidean norm. Then* $b_n > b_i$ *for all* $i = 1, \ldots, n-1$.

Intuitively, in the case of a small a, (2.73) being violated for some $c \neq 0$ means that all $x_1, \ldots, x_{n-1}$ lie approximately in the same hyperplane, and x_n is well outside it. The condition (2.73) holding for all $c \in \mathbb{R}^p \setminus \{0\}$ can be expressed by saying that the matrix $\sum_{i=1}^{n-1} x_i x_i' - x_n x_n' + aI$ is positive definite.

Proof First we assume $a = 0$ (so that ridge regression becomes least squares); an extension to $a \geq 0$ will be easy. In this case H_n is the projection matrix onto the column space $\mathcal{C} \subseteq \mathbb{R}^n$ of the overall data matrix X_n and $I_n - H_n$ is the projection matrix onto the orthogonal complement $\mathcal{C}^\perp$ of $\mathcal{C}$. We can have $b_n \leq b_i$ for $i < n$ (or even $b_n^2 \leq b_1^2 + \cdots + b_{n-1}^2$) only if the angle between $\mathcal{C}^\perp$ and the hyperplane $\mathbb{R}^{n-1} \times \{0\}$ is 45° or less; in other words, if the angle between $\mathcal{C}$ and that hyperplane is 45° or more; in other words, if there is an element $(c \cdot x_1, \ldots, c \cdot x_n)'$ of $\mathcal{C}$ such that its last coordinate is $c \cdot x_n = 1$ and its projection $(c \cdot x_1, \ldots, c \cdot x_{n-1})'$ onto the other coordinates has length at most 1.

To reduce the case $a > 0$ to $a = 0$ add the p dummy objects $\sqrt{a} e_i \in \mathbb{R}^p$, $i = 1, \ldots, p$, labelled by 0 at the beginning of the training set; here $e_1, \ldots, e_p$ is the standard basis of $\mathbb{R}^p$.

Lemma 2.17 *The case* $b_n \leq b_i$ *for* $i < n$ *is excluded from some* n *on almost surely under (A1)–(A4).*

Proof We will check that (2.73) holds from some n on. Let us set, without loss of generality, $a := 0$. Let $\Sigma_l := \frac{1}{l} \sum_{i=1}^{l} x_i x_i'$. Since $\lim_{l \to \infty} \Sigma_l = \Sigma$ a.s.,

$$|\lambda_{\min}(\Sigma_l) - \lambda_{\min}(\Sigma)| \to 0 \quad (l \to \infty) \quad \text{a.s.} \ ,$$

where $\lambda_{\min}(\cdot)$ is the smallest eigenvalue of the given matrix. Since $\|x_n\|^2 / n \to 0$ a.s.,

$$\frac{1}{n-1} \sum_{i=1}^{n-1} (c \cdot x_i)^2 = c' \Sigma_{n-1} c \geq \lambda_{\min}(\Sigma_{n-1}) \|c\|^2$$

$$> \frac{1}{2} \lambda_{\min}(\Sigma) \|c\|^2 > \frac{\|c\|^2 \|x_n\|^2}{n-1} \geq \frac{(c \cdot x_n)^2}{n-1}$$

for all $c \neq 0$ from some n on.

Proof Proper

As before, X stands for the data matrix X_{n-1} based on the first $n-1$ examples. A simple but tedious computation (whose details can be found in [40, Appendix A]) gives

$$t_i = \frac{a_i - a_n}{b_n - b_i} = \hat{y}_n + (y_i - \hat{y}_i) \frac{1 + g_n}{1 + g_i} \ , \tag{2.74}$$

where $g_i := x_i'(X'X + aI)^{-1}x_n$ (cf. (2.60)). The term $\hat{y}_n$ in (2.74) is the centre of the Bayesian prediction interval (2.61); it does not depend on i. We can see that

$$B^* - C^* = \sqrt{1 + g_n}\zeta_\alpha - (1 + g_n)V_{(k_n)} \tag{2.75}$$

(cf. (2.61)), where $\zeta_\alpha := \mathbf{z}_{\epsilon/2}\sigma$ (as defined after (2.72)), $k_n := \lceil \alpha n \rceil$ (cf. (2.43)), and $V_{(k_n)}$ is the k_nth order statistic in the series

$$V_i := \frac{r_i}{1 + g_i} \tag{2.76}$$

of residuals $r_i := y_i - \hat{y}_i$ adjusted by dividing by $1 + g_i$. The behaviour of the order statistics of residuals is well studied: see, e.g., the theorem in [43]. The presence of $1 + g_i$ complicates the situation, and so we first show that g_i is small with high probability.

Lemma 2.18 *Let $\eta_1, \eta_2, \ldots$ be a sequence of IID random variables with a finite second moment. Then $\max_{i=1,\ldots,n} |\eta_i| = o(n^{1/2})$ in probability (and even almost surely) as $n \to \infty$.*

Proof By the strong law of large numbers the sequence $\frac{1}{n}\sum_{i=1}^n \eta_i^2$ converges a.s. as $n \to \infty$, and so $\eta_n^2/n \to 0$ a.s. This implies that $\max_{i=1,\ldots,n} |\eta_i| = o(n^{1/2})$ a.s.

Corollary 2.19 *Under the conditions of the theorem, $\max_{i=1,\ldots,n} |g_i| = o(n^{-1/2})$ in probability.*

Proof Similarly to the proof of Lemma 2.17, we have, for almost all sequences $x_1, x_2, \ldots$,

$$\max_{i=1,\ldots,n} |g_i| \le \frac{\|x_n\| \max_{i=1,\ldots,n} \|x_i\|}{\lambda_{\min}(X'X + aI)} < 2\frac{\|x_n\| \max_{i=1,\ldots,n} \|x_i\|}{(n-1)\lambda_{\min}(\Sigma)}$$

from some n on. It remains to combine this with Lemma 2.18 and the fact that, by Assumption (A1), $\|x_n\|$ is uniformly bounded by a constant with high probability.

Corollary 2.20 *Under the conditions of the theorem, $n^{1/2}\left(r_{(k_n)} - V_{(k_n)}\right) \to 0$ in probability.*

Proof Suppose that, on the contrary, there are $\epsilon > 0$ and $\delta > 0$ such that $n^{1/2}\left|r_{(k_n)} - V_{(k_n)}\right| > \epsilon$ with probability at least δ for infinitely many n. Fix such ϵ and δ. Suppose, for concreteness, that, with probability at least δ for infinitely many n, we have $n^{1/2}\left(r_{(k_n)} - V_{(k_n)}\right) > \epsilon$, i.e., $V_{(k_n)} < r_{(k_n)} - \epsilon n^{-1/2}$ (the δ may have to be halved for that). The last inequality implies that $V_i < r_{(k_n)} - \epsilon n^{-1/2}$ for at least k_n values of i. By the definition (2.76) of V_i this in turn implies that $r_i < r_{(k_n)} - \epsilon n^{-1/2} + g_i r_{(k_n)}$ for at least k_n values of i. By Corollary 2.19, however, the last addend is less than $\epsilon n^{-1/2}$ with probability at least $1 - \delta$ from some n on (the fact that $r_{(k_n)}$ is bounded with high probability follows, e.g., from Lemma 2.21 below). This implies $r_{(k_n)} < r_{(k_n)}$ with positive probability from some n on, and this contradiction completes the proof.

The last (and most important) component of the proof is the following version of the theorem in [43], itself a version of the famous Bahadur representation theorem [13]. Details of the proof (under our assumptions) are spelled out in [40, Appendix B].

Lemma 2.21 ([43], Theorem) *Under the conditions of Theorem 2.7,*

$$n^{1/2}\left|\left(r_{(k_n)} - \zeta_\alpha\right) - \frac{\alpha - F_n(\zeta_\alpha)}{f(\zeta_\alpha)} + \mu'(\hat{w}_n - w)\right| \to 0 \qquad a.s.\,, \tag{2.77}$$

where F_n is the empirical distribution function of the noise $\xi_1, \ldots, \xi_{n-1}$ and $\hat{w}_n := (X'X + aI)^{-1}X'Y$ is the ridge regression estimate of w.

By (2.75), Corollary 2.19, and Slutsky's lemma (see, e.g., [361, Lemma 2.8]), it suffices to prove (2.72) with the left-hand side replaced by $n^{1/2}(V_{(k_n)} - \zeta_\alpha)$. Moreover, by Corollary 2.20 and Slutsky's lemma, it suffices to prove (2.72) with the left-hand side replaced by $n^{1/2}(r_{(k_n)} - \zeta_\alpha)$; this is what we will do.

Lemma 2.21 holds in the situation where w is a constant vector (the conditions of Theorem 2.7 allow the distribution of w to be degenerate). Let R be a Borel set in $(\mathbb{R}^p)^\infty$ such that (2.77) holds for all $(x_1, x_2, \dots) \in R$, where the "a.s." is now interpreted as "for almost all sequences $(\xi_1, \xi_2, \dots)$". By Lebesgue's dominated convergence theorem, it suffices to prove (2.72) with the left-hand side replaced by $n^{1/2}(r_{(k_n)} - \zeta_\alpha)$ for a fixed w and a fixed sequence $(x_1, x_2, \dots) \in R$. Therefore, we fix w and $(x_1, x_2, \dots) \in R$; the only remaining source of randomness is $(\xi_1, \xi_2, \dots)$. Finally, by the definition of the set R, it suffices to prove (2.72) with the left-hand side replaced by

$$n^{1/2} \frac{\alpha - F_n(\zeta_\alpha)}{f(\zeta_\alpha)} - n^{1/2} \mu'(\hat{w}_n - w) \,. \tag{2.78}$$

Without loss of generality we will assume that $\frac{1}{n} X_n' X_n \to \Sigma$ as $n \to \infty$ (this extra assumption about R will ensure that Lindeberg's condition is satisfied below).

Since $\mathbb{E}(\alpha - F_n(\zeta_\alpha)) = 0$ and

$$\operatorname{var}(\alpha - F_n(\zeta_\alpha)) = \frac{F(\zeta_\alpha)(1 - F(\zeta_\alpha))}{n-1} = \frac{\alpha(1-\alpha)}{n-1} \,,$$

where F is the distribution function of $\mathbf{N}(0, \sigma^2)$, we have

$$n^{1/2} \frac{\alpha - F_n(\zeta_\alpha)}{f(\zeta_\alpha)} \xrightarrow{\text{law}} \mathbf{N}\left(0, \frac{\alpha(1-\alpha)}{f^2(\zeta_\alpha)}\right) \quad (n \to \infty)$$

by the central limit theorem (in its simplest form).

Since $\hat{w}_n = (X'X + aI)^{-1} X'Y$ is the ridge regression estimate,

$$\begin{aligned} \mathbb{E}(\hat{w}_n - w) &= -a(X'X + aI)^{-1} w =: \Delta_n, \\ \operatorname{var}(\hat{w}_n) &= \sigma^2 (X'X + aI)^{-1} X'X (X'X + aI)^{-1} =: \Omega_n \,. \end{aligned}$$

Furthermore, as $n \to \infty$,

$$\begin{aligned} n^{1/2} \Delta_n &= -n^{-1/2} a \left(\frac{X'X}{n} + \frac{aI}{n} \right)^{-1} w \sim -n^{-1/2} a \Sigma^{-1} w \to 0 \,, \\ n\Omega_n &= \sigma^2 \left(\frac{X'X}{n} + \frac{aI}{n} \right)^{-1} \frac{X'X}{n} \left(\frac{X'X}{n} + \frac{aI}{n} \right)^{-1} \to \sigma^2 \Sigma^{-1} \,. \end{aligned}$$

This gives

$$n^{1/2} \mu'(\hat{w}_n - w) \xrightarrow{\text{law}} \mathbf{N}\left(0, \sigma^2 \mu' \Sigma^{-1} \mu\right) \quad (n \to \infty)$$

(the asymptotic, and even exact, normality is obvious from the formula for $\hat{w}_n$).

Let us now calculate the covariance between the two addends in (2.78):

$$\begin{aligned}
&\operatorname{cov}\left(n^{1/2}\frac{\alpha - F_n(\zeta_\alpha)}{f(\zeta_\alpha)}, -n^{1/2}\mu'(\hat{w}_n - w)\right) \\
&= \frac{n}{f(\zeta_\alpha)}\operatorname{cov}\left(F_n(\zeta_\alpha) - \alpha, \mu'(\hat{w}_n - w)\right) \\
&= \frac{n}{(n-1)f(\zeta_\alpha)}\sum_{i=1}^{n-1}\operatorname{cov}\left(1_{\xi_i \le \zeta_\alpha} - \alpha, \mu'(\hat{w}_n - w)\right) \\
&= \frac{n}{(n-1)f(\zeta_\alpha)}\sum_{i=1}^{n-1}\mathbb{E}\Big(\left(1_{\xi_i \le \zeta_\alpha} - \alpha\right)\mu'(X'X + aI)^{-1}X'\xi\Big) ,
\end{aligned}$$

where $\xi = (\xi_1, \ldots, \xi_{n-1})'$ and the last equality uses the decomposition $\hat{w}_n - w = \Delta_n + (X'X + aI)^{-1}X'\xi$ with the second addend having zero expected value. Since

$$\mathbb{E}1_{\xi_i \le \zeta_\alpha}\mu'(X'X + aI)^{-1}X'\xi = \sum_{j=1}^{n-1}\mathbb{E}1_{\xi_i \le \zeta_\alpha}A_j\xi_j = \mu_\alpha A_i ,$$

where

$$\begin{aligned}
A_j &:= \mu'(X'X + aI)^{-1}x_j, \quad j = 1, \ldots, n-1 , \\
\mu_\alpha &:= \mathbb{E}1_{\xi_i \le \zeta_\alpha}\xi_i = \int_{-\infty}^{\zeta_\alpha} x f(x)\mathrm{d}x ,
\end{aligned}$$

and an easy computation gives $\mu_\alpha = -\sigma^2 f(\zeta_\alpha)$, we have

$$\begin{aligned}
\operatorname{cov}\left(n^{1/2}\frac{\alpha - F_n(\zeta_\alpha)}{f(\zeta_\alpha)}, -n^{1/2}\mu'(\hat{w}_n - w)\right) &= \frac{n}{(n-1)f(\zeta_\alpha)}\sum_{i=1}^{n-1}\mu_\alpha A_i \\
&= -\sigma^2\frac{n}{(n-1)}\sum_{i=1}^{n-1}A_i = -\sigma^2\mu'\left(\frac{1}{n}X'X + \frac{a}{n}I\right)^{-1}\bar{x} \to -\sigma^2\mu'\Sigma^{-1}\mu
\end{aligned}$$

as $n \to \infty$, where $\bar{x}$ is the arithmetic mean of $x_1, \ldots, x_{n-1}$. Finally, this implies that (2.78) converges in law to

$$\mathbf{N}\left(0, \frac{\alpha(1-\alpha)}{f^2(\zeta_\alpha)} + \sigma^2\mu'\Sigma^{-1}\mu - 2\sigma^2\mu'\Sigma^{-1}\mu\right) = \mathbf{N}\left(0, \frac{\alpha(1-\alpha)}{f^2(\zeta_\alpha)} - \sigma^2\mu'\Sigma^{-1}\mu\right) ;$$

the asymptotic normality of (2.78) follows from the central limit theorem with Lindeberg's condition, which holds since (2.78) is a linear combination of the noise random variables $\xi_1, \ldots, \xi_{n-1}$ with coefficients whose maximum is $o(1)$ as $n \to \infty$ (this uses the assumption $\frac{1}{n}X_n'X_n \to \Sigma$ made earlier).

2.8.3 Proof of Theorem 2.10

Let $\xi_n^{(\epsilon)}$ be the random variables from the definition of the conservative validity of Γ, $\epsilon \in (0,1)$, $n \in \mathbb{N}$, and Q be a probability distribution on $\mathbf{Z}$. For each sequence of examples $(z_1, \dots, z_n) \in \mathbf{Z}^n$ let $f(z_1, \dots, z_n)$ be the following version of the conditional probability under $Q^\infty \times \mathbf{U}$ that $\xi_n^{(\epsilon)} = 1$ given $z_1, \dots, z_n$:

$$f(z_1, \dots, z_n) := \int \xi_n^\epsilon((z_1, \dots, z_n, \dots), \tau)\mathbf{U}(\mathrm{d}\tau) \; .$$

For each bag $B \in \mathbf{Z}^{(n)}$, let $f(B)$ be the arithmetic mean of $f(z_1, \dots, z_n)$ over all $n!$ orderings of B. We know that the expected value of $f(B)$ is ϵ under any Q^n, and this, by the completeness of the statistic that maps data sequences $(z_1, \dots, z_n)$ to bags $\wr z_1, \dots, z_n \int$ (see [222, Example 4.3.4]; since $\mathbf{Z}$ is standard Borel, it can as well be taken to be $\mathbb{R}$), implies that $f(B) = \epsilon$ for almost all (under any Q^∞) bags B.

Let us show that we can omit "almost". If there is a bag for which $f(B) \neq \epsilon$, defining Q to be the uniform probability distribution on the distinct elements of this bag leads to a contradiction: not Q^∞-almost all bags B satisfy $f(B) = \epsilon$.

Define $S(B, \epsilon)$ as the bag of elements z of B such that Γ makes an error at significance level ϵ at trial n when fed with the elements of B ordered in such a way that the nth example is z (since Γ is invariant, whether an error is made depends only on which element is last, not on the ordering of the first $n-1$ elements). It is clear that

$$\epsilon_1 \le \epsilon_2 \Longrightarrow S(B, \epsilon_1) \subseteq S(B, \epsilon_2)$$

(as Γ^ϵ is a nested family) and

$$\frac{|S(B, \epsilon)|}{n} \le \epsilon$$

(as $f(B) = \epsilon$). Therefore, the conformal predictor determined by the conformity measure

$$A(B, z) := \inf\{\epsilon : z \in S(B, \epsilon)\}$$

is at least as good as Γ.

2.9 Context

2.9.1 Exchangeability vs Randomness

In this book we establish various notions of validity for conformal prediction under two closely related assumptions, randomness and exchangeability. The relation becomes very simple in the case of infinite sequences, according to de Finetti's theorem (see Sect. A.5.1 in Appendix A): in wide generality, an exchangeable probability measure is a mixture of power probability measures Q^∞. For a finite horizon, the relation becomes less close.

According to Savage [304, Sect. 3.7] and Hewitt and Savage [158, Sect. 1], Haag (1924, [145]) seems to have been the first to discuss the concept of exchangeability; he also hinted at de Finetti's theorem but did not state it explicitly. De Finetti's theorem in the binary case was obtained by

Bruno de Finetti (1906–1985).

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him in 1930 [75] and independently by Khinchin in 1932 [190]; the extension to the general case (stated as the result about real-valued random variables) is due to de Finetti [76], and an abstract statement first appeared in [158].

In response to Hewitt and Savage's statement about Haag's role, Good [136, Chap. 3] drew attention to W. E. Johnson's role. In the final section, "Appendix on eduction", of his *Logic* [182], Johnson introduced exchangeability under the name of *permutability postulate* and used it in eduction, as we already mentioned in Chap. 1 (Sect. 1.3.4). See [2] for more information about Johnson.

Dale [68] reviews the early work on exchangeability, first of all, Johnson's, Haag's, and de Finetti's. He believes that "It would be unfair to single out any one of these three authors, at the expense of the others, as the father of exchangeability."

Relation between randomness and exchangeability for finite sequences has been explored by Diaconis and Freedman [83]. While an exchangeable distribution P on the examples $z_1, \dots, z_N$ is not necessarily close in the variation distance to a mixture of power distributions Q^N even for large N, the situation changes if P can be extended to an exchangeable distribution on $z_1, \dots, z_N, \dots, z_{N'}$ for $N' \gg N$: there is a mixture of power distributions at a variation distance of $N(N-1)/N'$, which can be made small by increasing N'. They give examples showing that this rate cannot be improved.

In the binary case, the difference between exchangeability and randomness for finite sequences will be discussed in Sects. 9.1.1 and 9.5.1. For a further discussion of exchangeability, see Sect. 11.6.3.

2.9.2 Conformal Prediction

Conformal predictors were first described in [400] (June 1999) and [303]. The independence of errors in the online mode was proved in [383].

The idea of a rudimentary conformal predictor based on Vapnik's support vector machine was described in [126] (January 1997) and originated at the discussions (mentioned in the preface to the first edition) between Gammerman, Vapnik, and Vovk in the summer of 1996. Having had worked for a long time on the algorithmic theory of randomness (the paper [396] being most relevant), Vovk realized that the fact that a small number of support vectors translates into confident predictions (cf. (13.6)) can be used for making hedged predictions. Let us consider the problem of classification ($|\mathbf{Y}|$ is finite and small). From the point of view of the algorithmic theory of randomness, we can make a confident prediction for the label y_n of the new object x_n given a training set $z_1, \dots, z_{n-1}$ if the algorithmic randomness deficiency is small for only one possible extension $(z_1, \dots, z_{n-1}, (x_n, y))$, $y \in \mathbf{Y}$; we can then output the corresponding y as a confident

prediction. (In the case of regression, $\mathbf{Y} = \mathbb{R}$, a confident prediction for x_n is possible if the algorithmic randomness deficiency is small for a narrow range of $y \in \mathbf{Y}$.)



Alexey Chervonenkis (1938–2014).

Used with permission of his son Mikhail Chervonenkis

Alexey Chervonenkis understood that a small number of support vectors translates into confident predictions already in June 1966, as we learned in 2004 from his recollections, later published in [407, Chap. 1]. See [407, Chap. 11] for a discussion of two approaches to machine learning born during the summer of 1966; in June Chervonenkis thought of using support vectors for expressing confidence (albeit formalizing it only as the inequality (13.6)), and in July he and Vapnik introduced the growth function and started developing VC theory.

Remark 2.22 The reader who is not familiar with the algorithmic theory of randomness (which is not used in this book outside the end-of-chapter remarks) can consult [205, 232, 238, 323, 415]. In the literature on algorithmic randomness the word "algorithmic" is often omitted, but we will always keep it, to avoid confusion with several other, unrelated, notions of randomness used in this book. (In particular, there are no obvious connections between algorithmic randomness and the assumption of statistical randomness.) The algorithmic notion of randomness formalizes the intuitive notion of typicalness: an object $\omega \in \Omega$ is regarded as typical of a probability distribution P on Ω (we will also say "under P" or "w.r. to P") if there is no reason to be surprised when learning that ω was drawn randomly from P. Using the notion of a universal Turing machine, it is possible to introduce the notion of *algorithmic randomness deficiency*, formalizing the degree of deviation from typicalness.

There are two very different approaches to defining algorithmic randomness deficiency: Martin-Löf's [238] and Levin's [229, 230] (the latter simplified by Gács [123] and in [396]). Kolmogorov's [204] original definition is a special case of Martin-Löf's, but becomes a special case of Levin's (as simplified in [396]) if the plain Kolmogorov complexity in it is replaced by prefix complexity. Martin-Löf's definition is more intuitive, being a universal version of the standard statistical notion of p-value, but Levin's definition often leads to more elegant mathematical results.

The paper [126] was based on Levin's definition, which made it difficult to understand. Conformal predictors, which appeared in [400] and [303], were the result of replacing Levin's definition of algorithmic randomness by Martin-Löf's definition in [126]. See [3, Sect. 7] for further connections between conformal prediction and the algorithmic theory of randomness.

After the notion of conformal predictor crystallized, the connection with the algorithmic theory of randomness started to disappear; in particular, in order to obtain the strongest possible results, we replaced the algorithmic notion of randomness with statistical tests. As we said earlier, in this book we hardly ever mention algorithmic theory of randomness outside the end-of-chapter remarks. This evolution does not look surprising: e.g., we argued in [395] that the algorithmic notions of randomness and complexity are powerful sources of intuition, but for stating mathematical

results in their strongest and most elegant form it is often necessary to "translate" them into a non-algorithmic form.

For the information on the many precursors of conformal prediction, see Sect. 13.3; Kei Takeuchi's definition is especially close to ours.

2.9.3 Two Equivalent Definitions of Nonconformity Measures

The first edition of this book [402] used the same definition of a nonconformity measure A, but it was applied to the definition of p-values and prediction sets in a different way. In particular, the values $A(\sigma, z)$ for examples $z \notin \sigma$ were also essential. Namely, we had the same definition via (2.22) of the prediction sets (2.21), but the formula (2.23) for computing nonconformity scores was

$$\begin{aligned} \alpha_i &:= A_n\big(\wr (x_1, y_1), \dots, (x_{i-1}, y_{i-1}), (x_{i+1}, y_{i+1}), \dots, (x_{n-1}, y_{n-1}), \\ &\qquad (x_n, y)\int, (x_i, y_i)\big), \qquad i = 1, \dots, n-1, \\ \alpha_n &:= A_n\big(\wr (x_1, y_1), \dots, (x_{n-1}, y_{n-1})\int, (x_n, y)\big) \end{aligned} \tag{2.79}$$

(we used the notation A_n for the restriction $A|_{\mathbf{Z}^n}$ of A to $\mathbf{Z}^n$). Of course, using such *deleted* nonconformity scores does not change the notion of a conformal predictor: they will lead to the same conformal prediction as our regular nonconformity scores computed from a conformity measure A if we compute the deleted nonconformity scores from the deleted nonconformity measure (2.35).

In Chap. 11 we will generalize conformal prediction replacing the exchangeability model by more general online compression models, which we already mentioned in Sect. 1.6.2. The equivalence between using regular and deleted nonconformity scores will hold for all online compression models used in this book, but in the first edition (Sect. 8.6) we considered a model, the Markov model, for which the equivalence is lost. The main definition of conformal prediction in the Markov model given in the first edition is in terms of deleted nonconformity scores, and it becomes impossible for regular nonconformity scores. However, that definition loses its efficiency in the case of asymmetric Markov chains: see Sect. 8.8 (subsection "Kolmogorov's modelling vs standard statistical modelling"). Alternative approaches to conformal prediction for Markov chains that do not use deleted nonconformity scores are discussed in Sect. 7.3 (subsection "Markov sequences") and Sect. 8.8 (the end of subsection "Kolmogorov's modelling vs standard statistical modelling") of the first edition.

The main advantages of our current approach (exemplified by (2.23)) are that it makes most of our formulas simpler (compare, e.g., (2.23) and (2.79)) and that it greatly simplifies the definition of "one-off structures" in Chap. 11 (Sect. 11.2.5). Some formulas, however, become more complicated. In such cases, we may use the deleted definition (2.35).

2.9.4 The Two Meanings of Conformity in Conformal Prediction

In the first edition [402] of this book we only used the first notion of conformity discussed in Sect. 2.2.9, conformity to a bag. The analogue of conformalized ridge regression for the symmetric nonconformity measure (2.30) in place of the asymmetric conformity measures (2.33) and (2.34) was called the ridge regression confidence machine. The asymmetric approach, which we interpret

in terms of conformity to a property, was proposed in [40] to facilitate theoretical analysis, but it also simplifies prediction algorithms. Asymmetric definitions such as (2.33) are widely used in conformal predictive distributions, which will be the topic of Chap. 7.

2.9.5 Examples of Nonconformity Measures

The main examples of nonconformity measures that we considered in this chapter (concentrating on the case of regression) are provided by the methods of nearest neighbours, least squares, and ridge regression. The origins of the nearest neighbours method are lost in the mists of time [275], but early modern references include Fix and Hodges's 1951 paper [113] (reprinted in 1989 with a commentary [333]) and [59] (with the method being used by Peter Hart and Charles Cole in 1966 [58]).

The least squares procedure was invented independently by Gauss and Legendre and first published by Legendre [221] in 1805 (for details, see, e.g., Plackett [279] and Stigler [342, 343]). The term "hat matrix" was introduced by John W. Tukey (see [163]). The ridge regression procedure was first described in detail by Arthur E. Hoerl and Robert W. Kennard [168, 169] in 1970. The idea came from Hoerl's [165] ridge analysis, a method of examining high-dimensional quadratic response surfaces (for details, see Hoerl [167]). The link between ridge analysis and ridge regression is provided by Hoerl's 1962 paper [166].

Deleted residuals are also known as PRESS and predicted residuals, and the nonconformity scores (2.47) differ from "internally studentized residuals" only by a factor that does not affect the conformal predictor's output. We did not consider "externally studentized residuals"; for details and history, see, e.g., [54, Sect. 2.2.1].

The CRR was introduced in [40], as we already mentioned, whereas its predecessor, ridge regression confidence machine (RRCM), was developed by Ilia Nouretdinov and published as [264]. For the pseudocode of various implementations of the RRCM, see the first edition of this book (Sect. 2.3) and Appendix B of the arXiv version of [405].

The conformalized version of the lasso was derived by Lei [224]. This result is much more sophisticated than our simple procedure, Algorithm 2.4, based on ridge regression.

An important advance is Romano et al.'s [293] conformalized quantile regression, which employs classical quantile regression to develop a conformal predictor that is adaptive to heteroscedasticity. Another approach to making conformal prediction adaptive to heteroscedasticity is Chernozhukov et al.'s [49] distributional conformal prediction.

2.9.6 Kernel Methods

Kernel methods are a powerful tool for modifying nonconformity measures and making them more versatile. They have their origins in the Hilbert–Schmidt theory of integral equations (see [249]). The fundamental fact that each symmetric nonnegative definite function has representation (2.54) can be proved by many different methods: see, e.g., Mercer [249] (that paper, however, proves a slightly different result, "Mercer's theorem", about continuous kernels and an integral analogue of condition (2.58)) and Aronszajn [7] (Aronszajn's proof is based on Moore's idea; it is reproduced in Wahba [416]). For more recent expositions, see, e.g., [67, 308, 322, 341]. In the first edition [402, p. 37] we gave a simple and standard probabilistic proof.

There are several approaches to kernel ridge regression; the three main ones appear to be the following:

- the approach adopted in this book: the objects are mapped to an arbitrary (not necessarily functional or separable) Hilbert space, and the prediction rule is chosen from among the continuous linear functionals on that space; the main Eq. (2.53) can be obtained, for example, using the Lagrange method analogously to Vapnik's [366] derivation of SVM (see [302]); the approach based on the equality (2.51) is standard in statistics;
- the approach based on functional Hilbert spaces with bounded evaluation functionals (called *reproducing kernel Hilbert spaces*; the prediction rule is chosen from among the elements of such a space; see, e.g., Wahba [416]);
- the approach based on Gaussian processes: one assumes that the labels y_i are obtained from zero-mean normal random variables with covariances $\mathrm{cov}(y_i, y_j)$ defined in terms of $\mathcal{K}(x_i, x_j)$; (2.53) can then be obtained as the expected value of the x's label. In geostatistics this approach is known as kriging; for further details, see Cressie [66] and Sect. 13.7.4.

An especially important class of kernels is formed by the *universal kernels* $\mathcal{K}$ defined to be continuous kernels such that the finite linear combinations $\sum_{i=1}^{n} c_i K(\cdot, x_i)$ are dense (in the maximum norm) in the set of all continuous functions on any compact subset of the object space $\mathbf{X}$, which is assumed to be a Hausdorff topological space. For details, see [250]. In the case of a compact $\mathbf{X}$ the definition goes back to Steinwart [340]; see also [341, Sect. 4.6]. If we consider the right-hand side of (2.53) as function of the test object x (remember that k_n is defined via x), this function can approximate any continuous function of x on any compact, provided $\mathcal{K}$ is a universal kernel. This makes the location of the prediction interval produced by CRR based on a universal kernel fully adaptive; its length, however, will be less adaptive. We will further discuss universal kernels in the context of conformal prediction, namely conformal predictive distributions, in Chap. 7.

Formula (2.56), which we used for the fast updating of the inverse matrix in the kernel CRR, may have been first explicitly given by Banachiewicz [16, 17]; further references and history can be found in [155]. There are similar updating formulas (going back to Gauss [128] and also reviewed in [155]) that could be used in the case of CRR, but the need for speeding up computations is less pressing for CRR since the matrix to be inverted is of size $p \times p$, which we assumed constant.

2.9.7 Burnaev–Wasserman Programme

In Sect. 2.5 we follow Burnaev and Vovk [40]. We state the main result asymptotically, although non-asymptotic results are definitely feasible. In fact, the main interest of strong parametric assumptions as soft models is in the possibility of non-asymptotic results and, even more importantly, in practical efficiency (non-asymptotic, of course).

Whereas in the case of the Burnaev–Wasserman programme asymptotic results serve as a means to express interesting phenomena in a simple way and are by no means indispensable, some of the results in the following chapters are infinitary in a very essential way, similarly to Stone's [347] result discussed in Sect. 1.5.1. Such intrinsically infinitary results include the existence of universal prediction algorithms in the context of classification (Sect. 3.2 of the first edition [402] of this book), probabilistic classification (Sect. 6.3), and probabilistic regression (Sect. 7.6).

2.9.8 Completeness Results

A version (more sophisticated but less precise, involving arbitrary constants) of Proposition 2.9 and Theorem 2.10 was stated and proved in [265]. An analogous result was stated by Takeuchi for his version of conformal predictors.

For a generalization of Theorem 2.10 to online compression models (introduced in Sect. 1.6.2 and defined in Part IV) different from exchangeability, see [405, Theorem 2].

Our results in Sect. 2.6 can be interpreted in terms of complete classes of decision procedures, in the statistical terminology [222, 420], or Pareto frontiers, in the economic and engineering terminology. Strictly speaking, the definitions of both complete classes and Pareto frontiers also need an irreflexive relation of strict domination, alongside the reflexive relation of domination. To simplify exposition, we did not introduce the former, but we can still roughly summarize Sect. 2.6 as the conformal predictors forming a complete class of valid confidence predictors.